

(12) **Patent Application Publication**
Marks

(10) **Pub. No.: US 2025/0276429 A1**
(43) **Pub. Date: Sep. 4, 2025**

(60) Provisional application No. 62/895,475, filed on Sep. 3, 2019, provisional application No. 62/843,553, filed on May 5, 2019.

Publication Classification

(51) **Int. Cl.**
B25C 5/11 (2006.01)

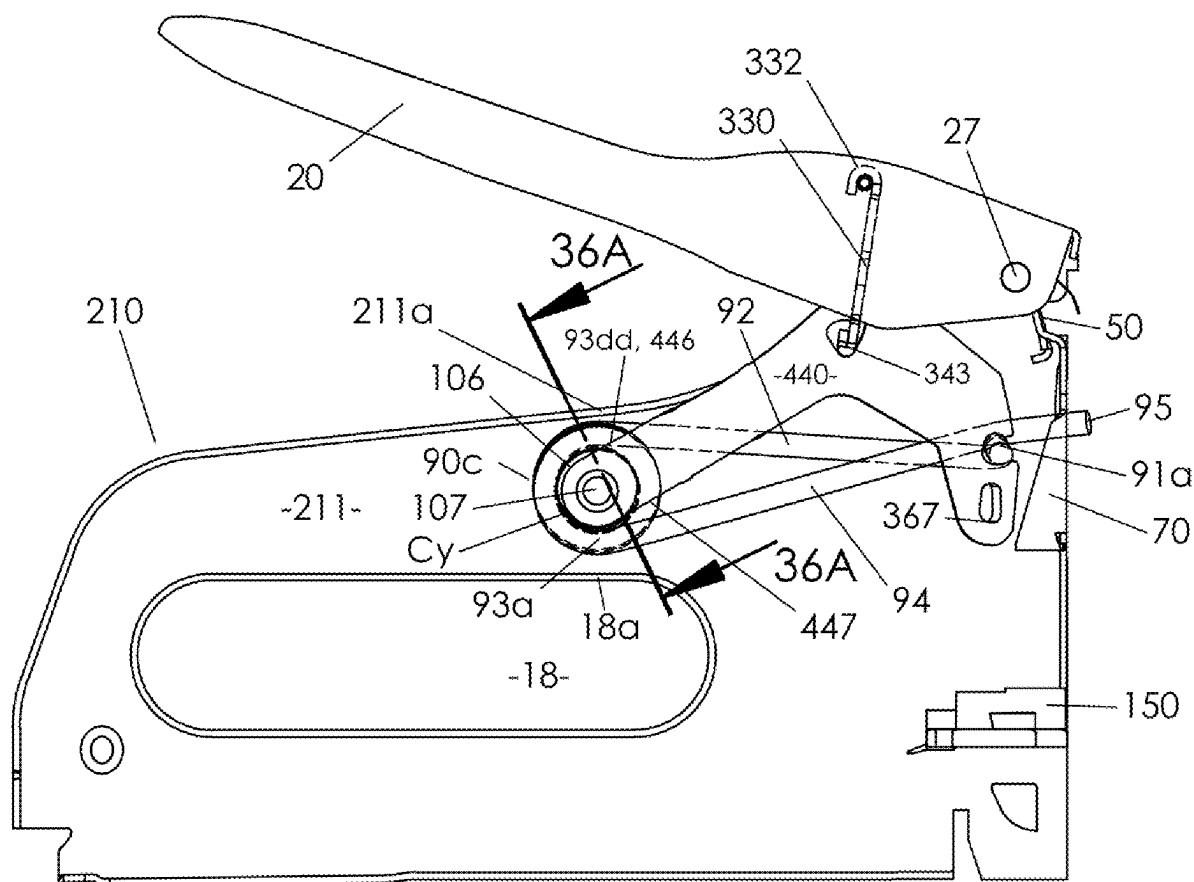
(52) **U.S. Cl.**
CPC *B25C 5/11* (2013.01)

(57) **ABSTRACT**

A spring energized fastening tool with compact, rigid, low friction working elements is disclosed. A torsion power spring includes forward extending arms with the arms pressing each other proximate a front distal end of the spring. A cantilevered lever links to a handle and engages the spring adjacent to the striker. A bottom loading staple track unlatches and opens through a simple pulling-out action. Structures are provided to enable fitment with a formed sheet metal handle and housing. The fastening tool is particularly simple to assemble, powerful, and of low operating effort.

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/657,739, filed on May 7, 2024, now Pat. No. 12,325,115, which is a continuation of application No. 17/942,098, filed on Sep. 9, 2022, now Pat. No. 11,981,010, which is a continuation of application No. 16/858,621, filed on Apr. 25, 2020, now Pat. No. 11,472,014.



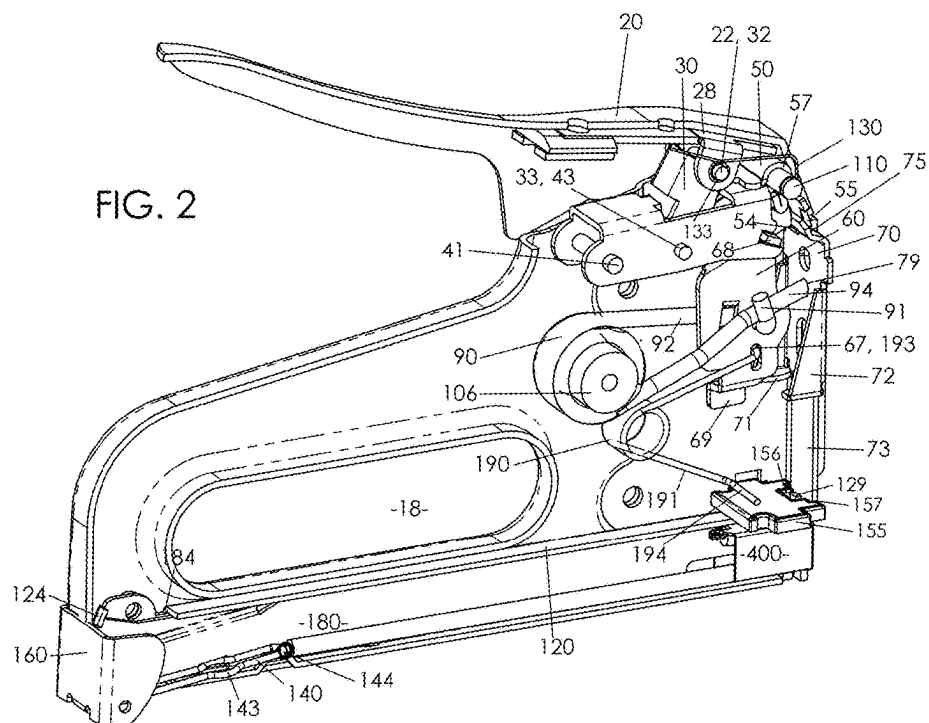
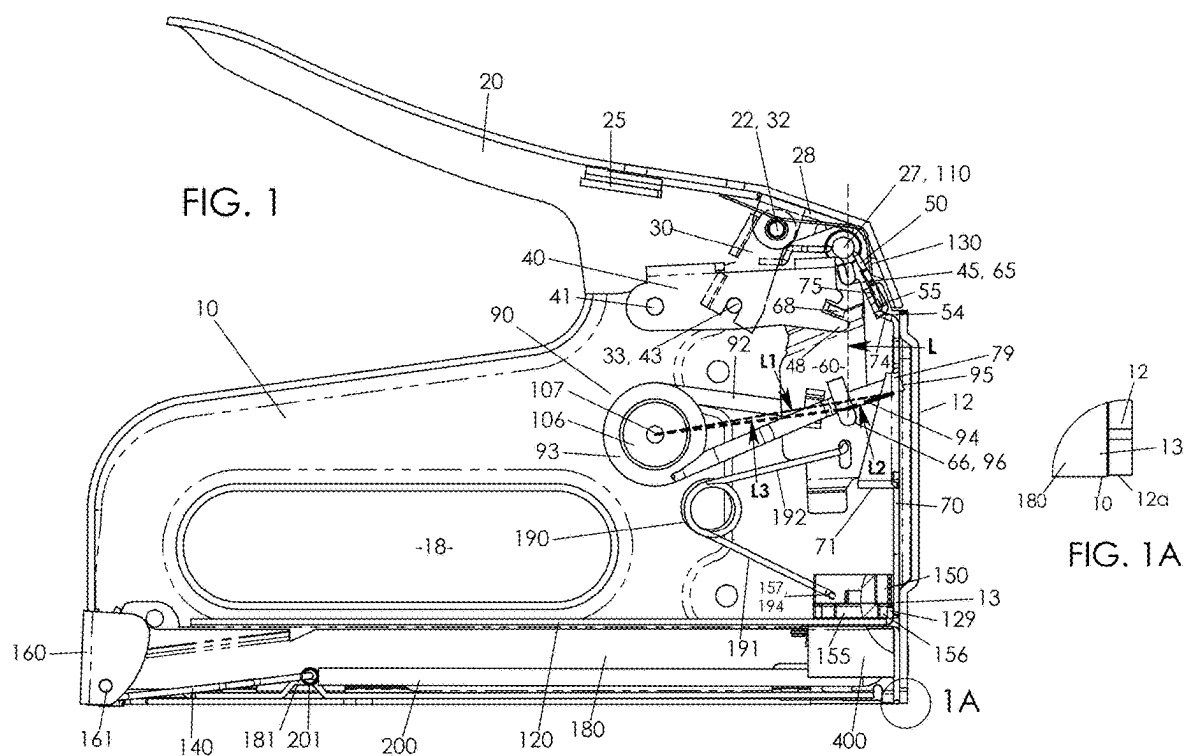


FIG. 3

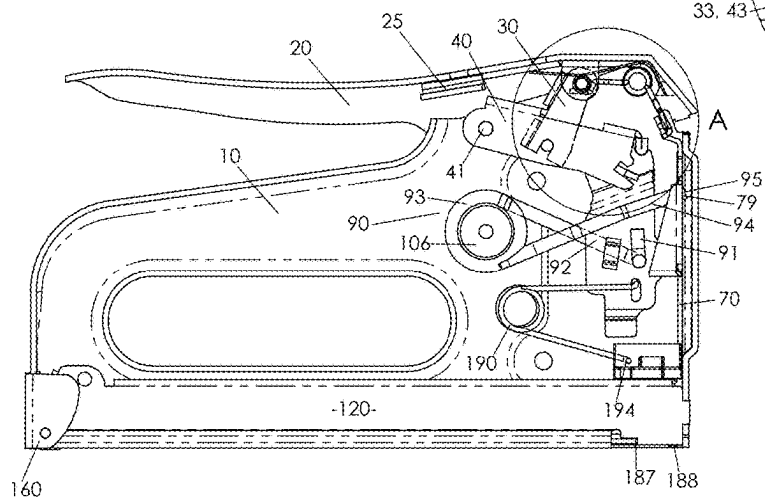


FIG. 4

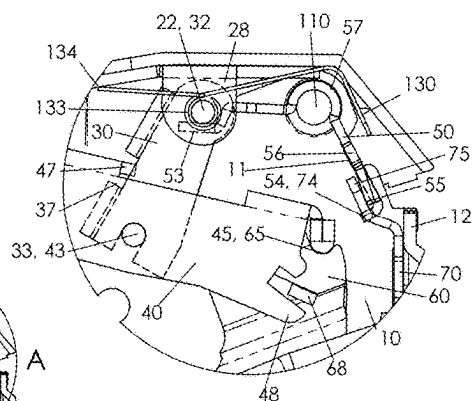
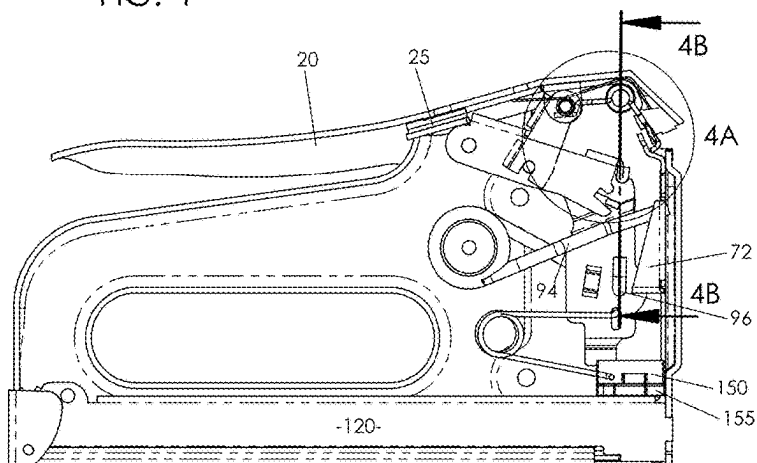


FIG. 3A

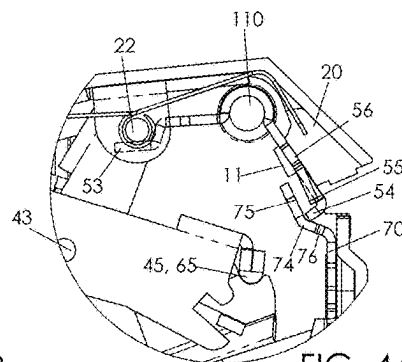


FIG. 4A

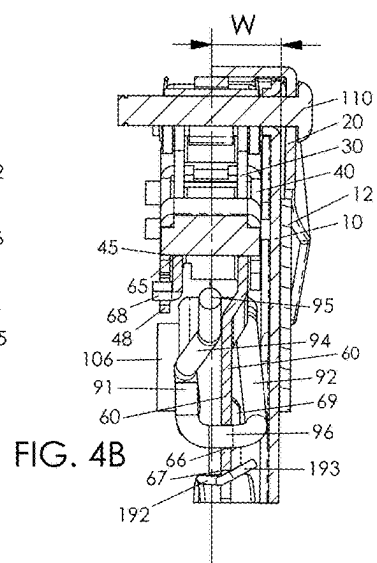


FIG. 4B

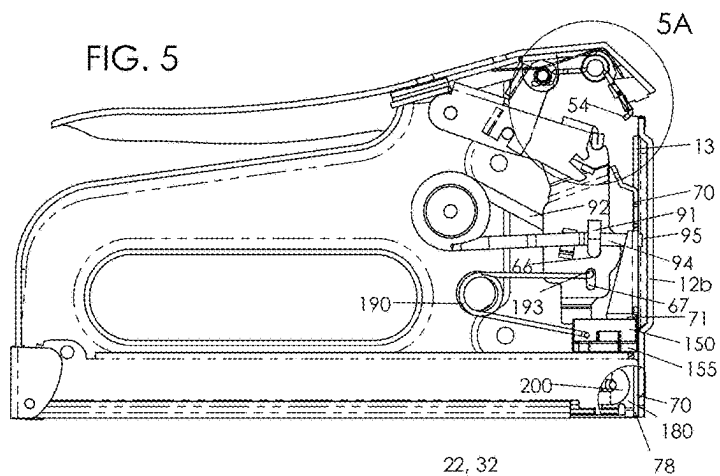


FIG. 5

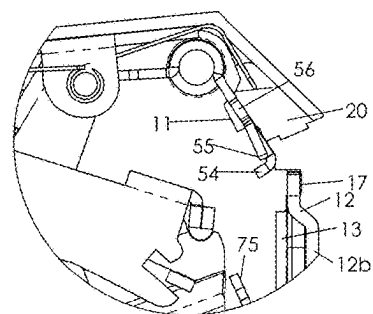


FIG. 5A

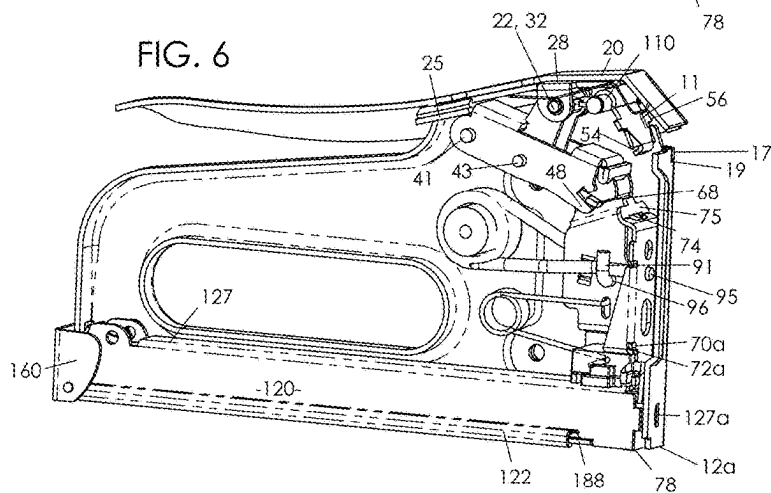


FIG. 6

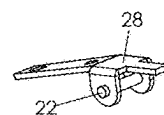


FIG. 7

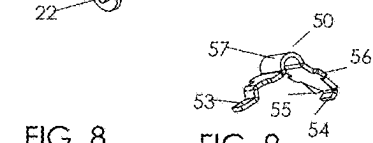


FIG. 9

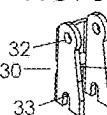


FIG. 8

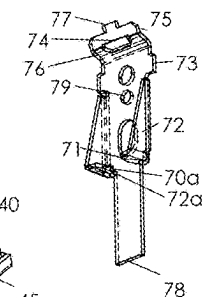


FIG. 11

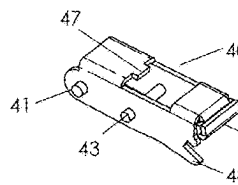


FIG. 10

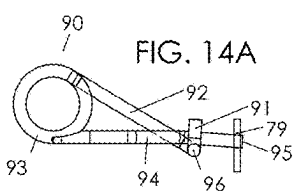


FIG. 14A

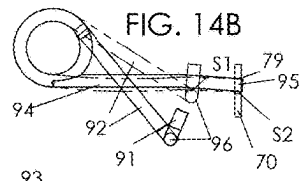


FIG. 14B

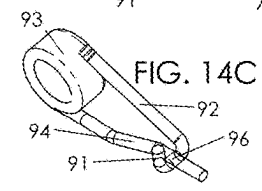


FIG. 14C

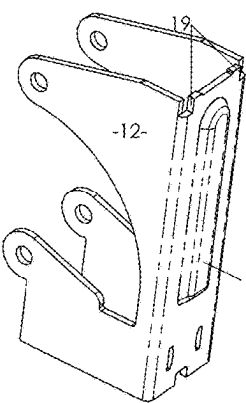


FIG. 13

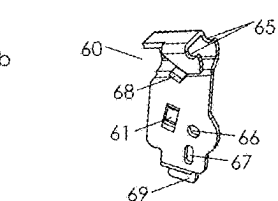


FIG. 12

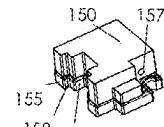


FIG. 15

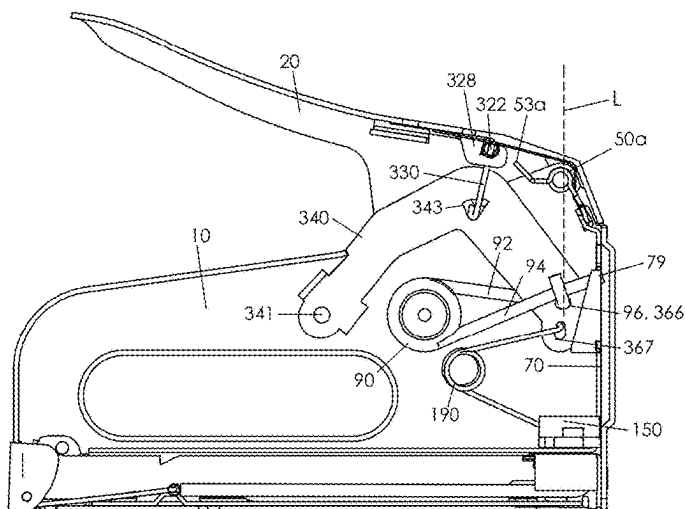


FIG. 16

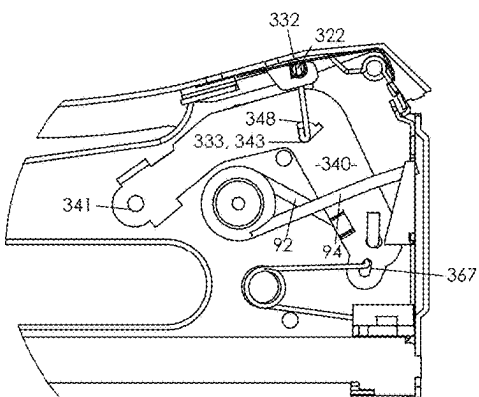


FIG. 17

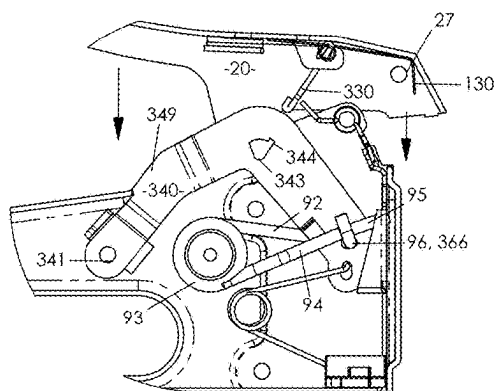


FIG. 18

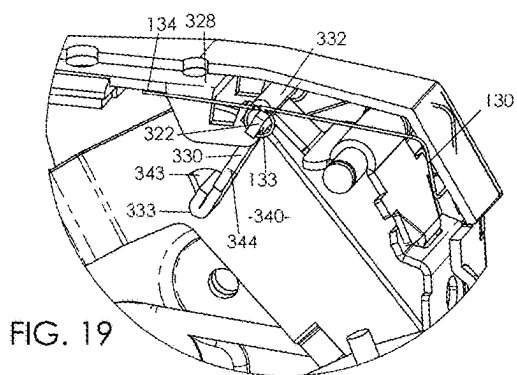


FIG. 19



FIG. 20

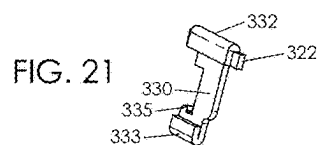


FIG. 21

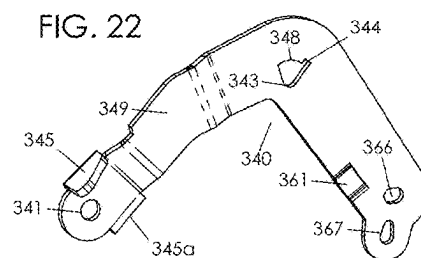


FIG. 22

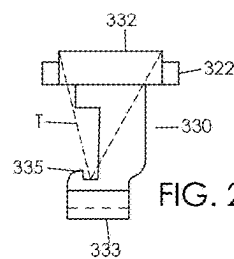
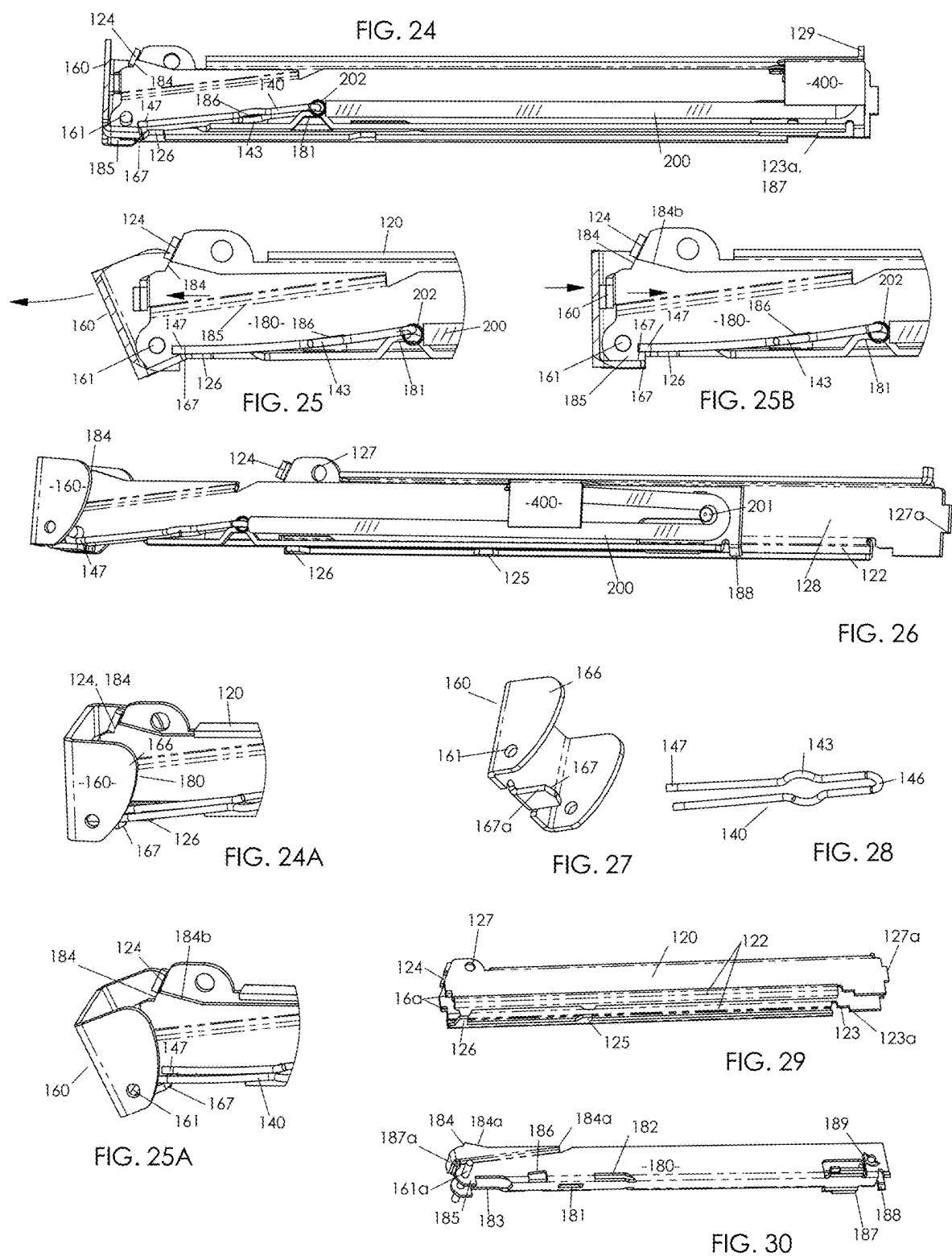


FIG. 23



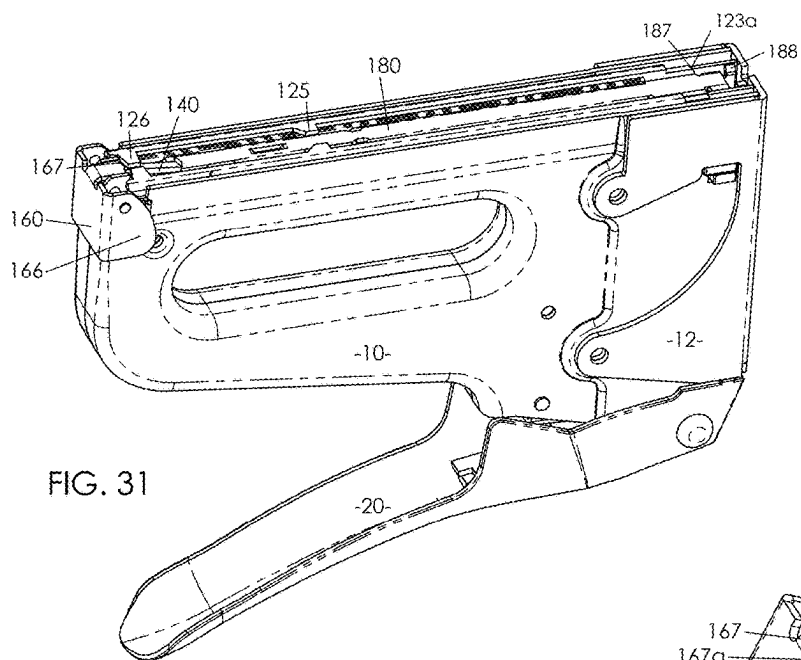


FIG. 31

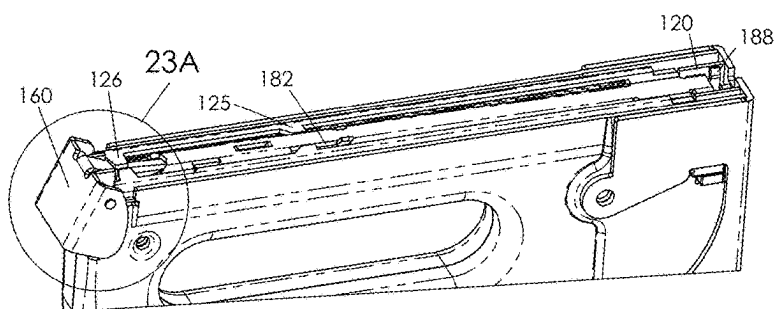


FIG. 32

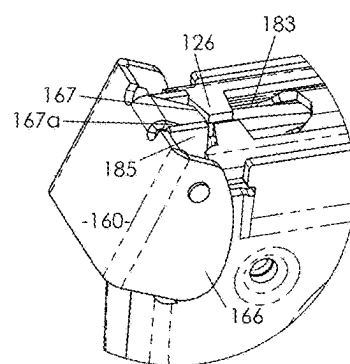


FIG. 32A

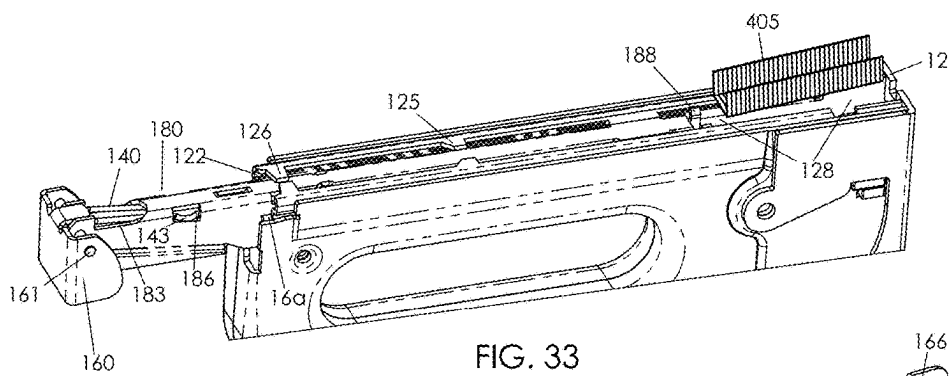


FIG. 33

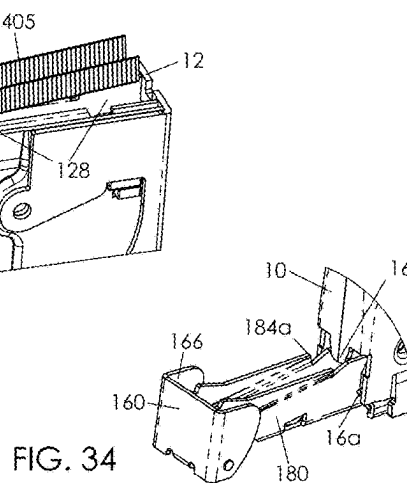


FIG. 34

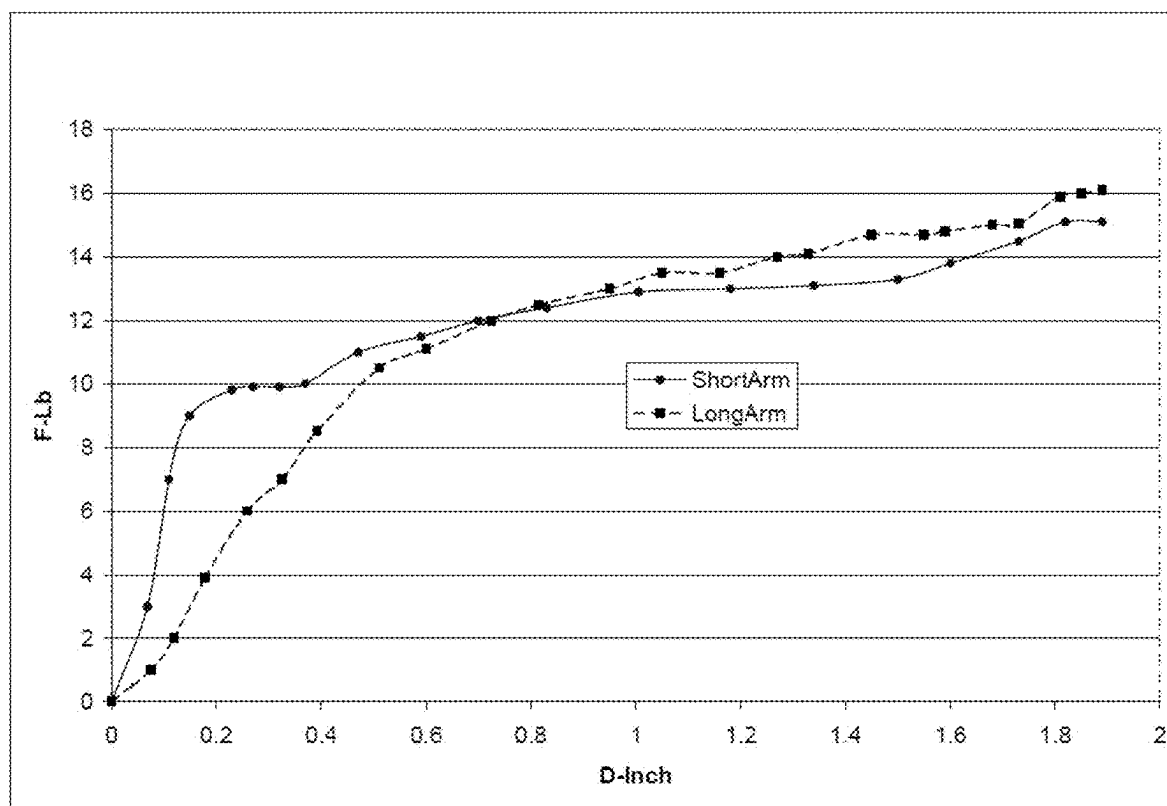


FIG. 35

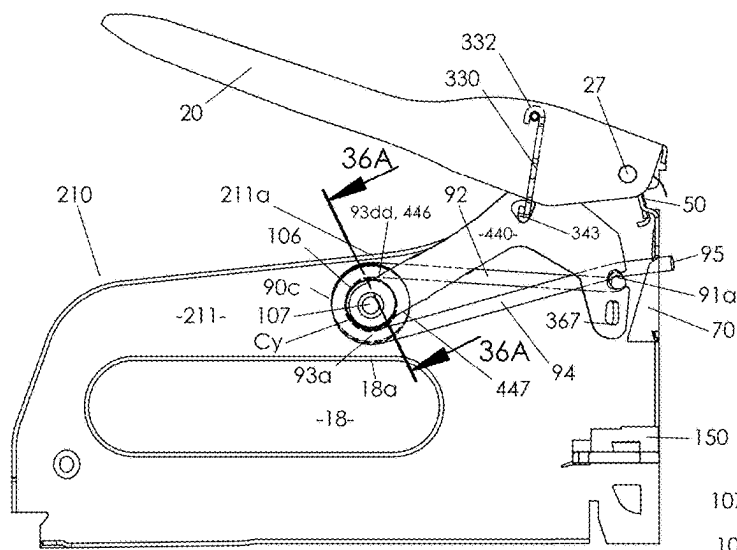


FIG. 36

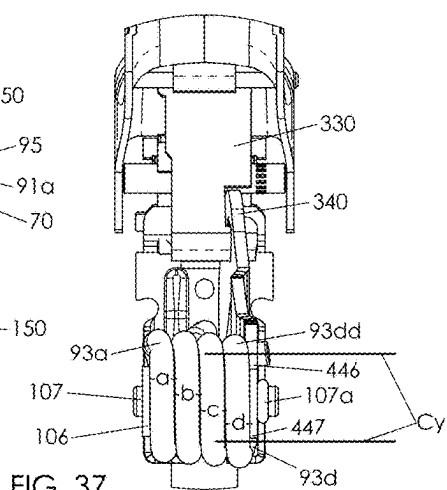


FIG. 37

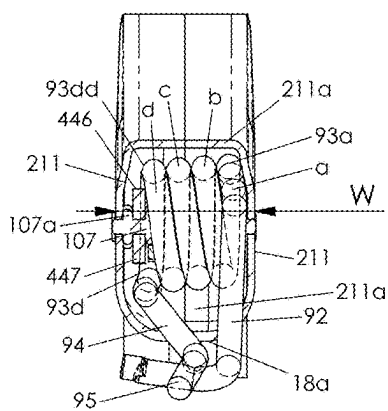


FIG. 36A

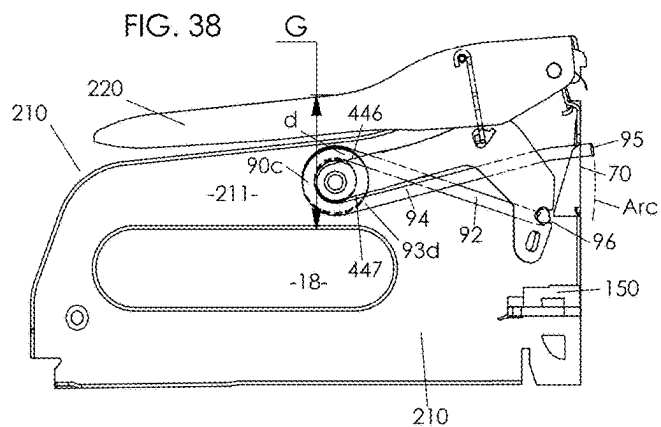


FIG. 38

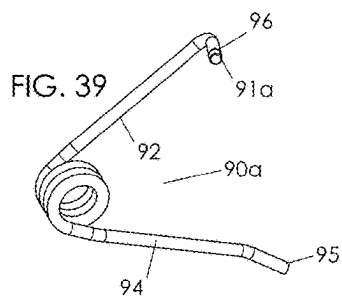


FIG. 39

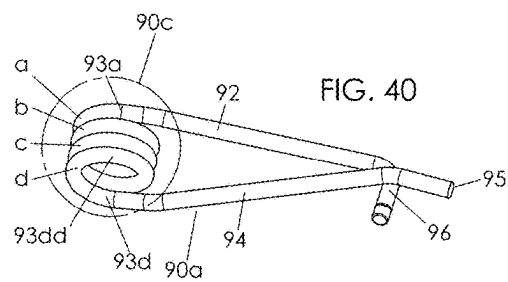


FIG. 40

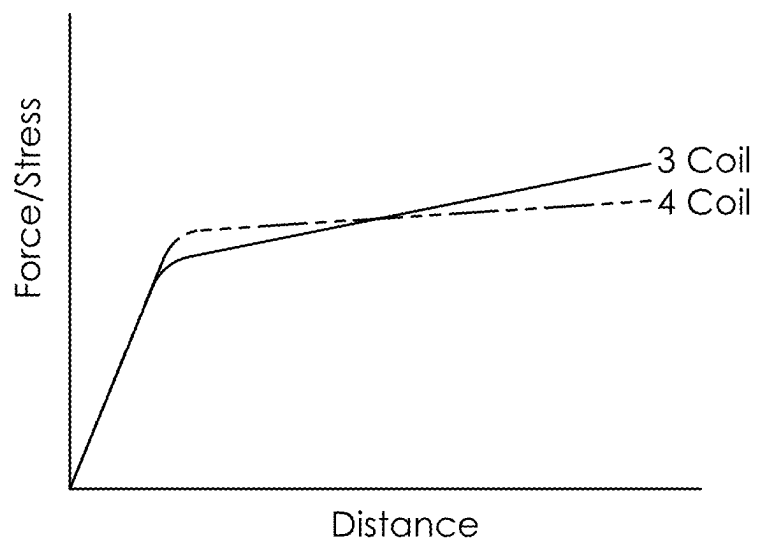


FIG. 41

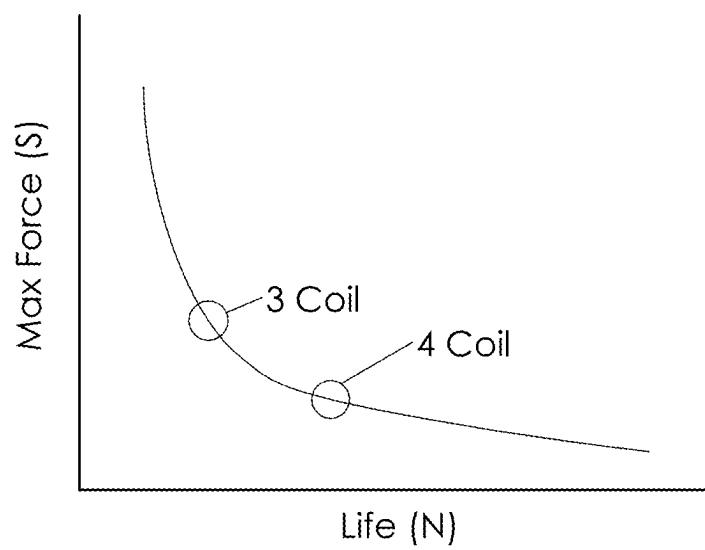


FIG. 42

HIGH EFFICIENCY TORSION SPRING TACKER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation-In-Part (CIP) of co-pending application Ser. No. 18/657,739, filed May 7, 2024, which is a continuation of application Ser. No. 17/942,098, filed Sep. 9, 2022, now U.S. Pat. No. 11,981,010, which is a continuation of application Ser. No. 16/858,621, filed Apr. 25, 2020, now U.S. Pat. No. 11,472,014, which claims priority from provisional application No. 62/895,475, filed Sep. 3, 2019, and from provisional application No. 62/843,553, filed May 5, 2019, the contents of all of which are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to spring energized tackers. More precisely, the present invention relates to a tacker with improved efficiency of assembly and operation.

BACKGROUND

[0003] Staple gun tackers and the like with energy storage via a power spring are known. A spring is deflected to store energy for sudden release to impact and drive a fastener into a work piece. Most commonly associated with manually operated hand tools such as a staple gun, a power spring based driving tool may also operate with a motorized system. A power spring may include a compression type, elongated bar, or torsion wire spring. With manual staple guns, a tool housing may include formed sheet metal, die cast, or resin molded. The sheet metal construction has most often been associated with compression springs and, less often, bar springs. One example of a sheet metal bodied staple gun is the T-50 brand of tacker, while many other such tackers are also known. Torsion springs are generally associated with molded or die cast housings; these are effective for providing the supports and guides for operating torsion springs.

[0004] The various springs may be used in a low start tacker, wherein the striker starts an operating cycle from a normal rest position in front of the staple or fastener track, and a high start where the striker normally rests above the staple track to start an operating cycle. In either case, there must be a release system to suddenly release the striker to instantly move down under the spring bias to eject a fastener. It is common that the release for one or both are imprecise and a source of force-adding friction.

[0005] A guide track for staples or fasteners is located along a bottom of the tool. Staples may be inserted from the rear or at the bottom among other known arrangements. Rear loading designs are prone to jamming since the staples cannot be easily accessed near the track front where jams may occur. Bottom loading exposes the full staple storage area for access as the track slide out rearward. A track pull with a latching structure is required to hold the track in its operative position. Such latches can be unwieldy and require aesthetic compromise.

SUMMARY OF THE INVENTION

[0006] In various preferred embodiments, the present invention is directed to a spring energized fastening tool with compact, low friction working elements. In the pre-

ferred high start embodiment, a torsion power spring includes at least two forward extending arms with the arms pressing each other proximate a front distal end of the spring. One embodiment has a rigid and movable four bar assembly that links the handle to the power spring and deflects the spring to separate and deflect the arms immediately upon pressing the handle. A further embodiment has a cantilevered lever engaging the spring adjacent to the striker. A release link preferably nests within a front portion of the handle whereby the release moves directly with the handle about a common pivot hinge during a release portion of the handle stroke. This structure provides reliable and repeatable release action.

[0007] Various preferred structures are provided to enable fitment with a formed sheet metal handle and housing. The illustrated structures are compatible to fit within the confines of a standard T-50 type tacker, for example, while also being well suited to other sheet metal, molded and die cast bodied tackers. As fitted, the fastening tool is particularly simple to assemble, powerful, and of low operating effort.

[0008] In the preferred embodiment, a bottom loading staple track is compatible with a sheet metal housing among other housing structures. The track unlatches and opens through a simple pulling-out action.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a partially cross-sectioned, side elevational view of a fastening tool in a rest condition according to one embodiment.

[0010] FIG. 1A is a detail view of FIG. 1 showing a lower front corner area.

[0011] FIG. 2 is a rear, top perspective view of the fastening tool of FIG. 1.

[0012] FIG. 3 is the tool of FIG. 1 in a pressed condition.

[0013] FIG. 3A is a detail view of a top front area of the tool of FIG. 3.

[0014] FIG. 4 is the tool of FIG. 1 in a pre-release condition.

[0015] FIG. 4A is a detail view of a top front area of the tool of FIG. 4.

[0016] FIG. 4B is a partial transverse cross-sectional view of a front area of the tool of FIG. 4.

[0017] FIG. 5 is the tool of FIG. 1 in a released condition.

[0018] FIG. 5A is a detail view of a top front area of the tool of FIG. 5.

[0019] FIG. 6 is a front perspective view of the tool of FIG. 5.

[0020] FIG. 7 is a front top perspective view of a handle link pivot support.

[0021] FIG. 8 is a front perspective view of a handle to lever link.

[0022] FIG. 9 is a rear perspective view of a release latch.

[0023] FIG. 10 is a top front perspective view of a lever.

[0024] FIG. 11 is a rear bottom perspective view of a striker.

[0025] FIG. 12 is a front top perspective view of a link bar.

[0026] FIG. 13 is a top front perspective view of a front cover.

[0027] FIG. 14A is a side elevational view of a power spring in a rest condition.

[0028] FIG. 14B is the spring of FIG. 14A with the spring partly deflected in phantom and the spring in a pressed condition.

[0029] FIG. 14C is a top perspective view of the spring of FIG. 14A.

[0030] FIG. 15 is a top, front perspective view of an absorber assembly.

[0031] FIG. 16 is a side elevational view of a fastening tool in a rest condition showing operative parts according to an alternative embodiment.

[0032] FIG. 17 is a cropped side elevational view of the tool of FIG. 16 in a pre-release condition.

[0033] FIG. 18 shows an assembly step of an upper handle sub-assembly to a lower tacker structure.

[0034] FIG. 19 is a detail view in perspective showing a handle and lever linkage during an assembly step.

[0035] FIG. 20 is a rear top perspective view of a rear handle link pivot support according to the alternative embodiment.

[0036] FIG. 21 is rear perspective view of a handle to lever link according to the alternative embodiment.

[0037] FIG. 22 is a side, rear perspective view of a lever according to the alternative embodiment.

[0038] FIG. 23 is a rear elevational view of the link of FIG. 21.

[0039] FIG. 24 is a side, bottom perspective view, partly in cross-section, of a track chamber subassembly.

[0040] FIG. 24A is a top, side perspective detail view of the subassembly of FIG. 24.

[0041] FIG. 25 is a rear detail view of the subassembly of FIG. 24 with the track in a de-latched condition and moving to open.

[0042] FIG. 25A is a top, side perspective detail view of the subassembly of FIG. 25.

[0043] FIG. 25B is the view of FIG. 25A with the track moving to the closed position.

[0044] FIG. 26 is a side, front perspective view of the subassembly of FIG. 24 with the track pulled out for staple loading.

[0045] FIG. 27 is a bottom, front perspective view of a track pull.

[0046] FIG. 28 is a bottom front perspective view of a track pull bias spring or latch spring.

[0047] FIG. 29 is a side, bottom perspective view of track guide chamber.

[0048] FIG. 30 is a side, bottom perspective view of a staple track.

[0049] FIG. 31 is a bottom, side perspective view of a tacker inverted in position in preparation for bottom loading of staples and fasteners, with the track in its closed operative position.

[0050] FIG. 32 is a cropped view of the tacker of FIG. 31 with the track pull de-latched.

[0051] FIG. 32A is a detail view of the tacker of FIG. 32.

[0052] FIG. 33 is the tacker of FIG. 32 with the track partly opened to expose a staple loading chamber.

[0053] FIG. 34 is a detailed top rear perspective view of the tacker of FIG. 33, in the upright position.

[0054] FIG. 35 is a handle force F (y axis) versus travel distance D (x axis) plot illustrating the performance advantages of a rigid handle-to-spring linkage.

[0055] FIG. 36 is a side elevational view of a fastening tool with internal elements exposed upon a left housing body, in a rest condition according to a further embodiment.

[0056] FIG. 36A is a partial section rearward view of the fastening tool of FIG. 36 near a central location of the spring coil.

[0057] FIG. 37 is a detailed rear elevational view of the fastening tool of FIG. 36 with the housing removed for clarity.

[0058] FIG. 38 is the fastening tool of FIG. 36, in a pre-release condition.

[0059] FIG. 39 is a perspective view of a power spring in a free condition.

[0060] FIG. 40 is the spring of FIG. 39 in a rest or preloaded condition.

[0061] FIG. 41 is a schematic comparative force distance plot.

[0062] FIG. 42 is an exemplary schematic S-N (stress/cycle) curve for a spring.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0063] The present invention is directed to a compact, efficient spring energized tacker that may operate and be fitted within a formed sheet housing body or like standardized body. The drawings show a preferred embodiment tacker that has a body sized and shaped similarly to a known commercially available tacker operable with T-50 style staples up to $\frac{1}{2}$ " or $\frac{3}{16}$ " long. However, the features of the present invention function with tackers of other shapes, sizes, and constructions including molded resin and die cast. For example, one or both of housing 10 and handle 20 may include sheet metal, molded resin, and/or die cast metal. In describing a tacker, such term may include staple guns, nail guns and equivalent fastening tools whether motorized or manually powered to energize a power spring.

[0064] In the preferred embodiment tacker of FIG. 1, for example, the length of the tool from the rear end to front end is $7\frac{1}{4}$ inches long. In FIG. 4B, housing 10 totals approximately 0.9 inch wide at the dimension W (W being doubled from approximately 0.45 inch to include the opposed housing side that is not shown). Other sizes, shapes and dimensions of the housing, handle, and other operating parts are contemplated.

[0065] In the assembly drawings FIG. 1 to FIG. 6, a right side of the housing is removed and handle 20 is depicted in cross-section to show internal components. Housing 10 has a front (right side of FIG. 1), a rear, a top, and a bottom. FIG. 1 shows a rest condition of the tacker. Handle 20 is in an upper position above housing 10, and it is pivotally attached to housing 10 at a handle/housing pivot, here hinge pin 110 near the top of the housing. At the bottom of housing 10 is a staple track 180 that supports staples biased forward by spring-driven pusher 400. Handle link pivot support 28 includes pivot hinge 22. Link 30 has pivot 32 fitted to hinge 22 defining an upper end or equivalent location of a linkage assembly. A lower end of link 30 includes slot 33 to engage hinge 43 of lever 40. See also FIGS. 7 to 15 for the individual components. Lever 40 includes pivot tab 45 to engage groove 65 of link bar 60. Link bar 60 engages pivot, hinge pin or hinge element 96 of power spring 90 at link bar hole 66. Hole 66 may define a lower end or equivalent structure of the linkage assembly that begins at hinge 22. The linkage lower end is below and substantially forward from the linkage upper end. As seen in FIG. 1, imaginary vertical line L between the pivot structures of hinge pin 110 and element 96 is well forward of hinge 22; as illustrated, line L is close to the forward structure or blade of striker 70. [0066] As seen in FIGS. 2 and 3, the power spring 90 pivots about mandrel 106. Power spring arm 94 extends

from spring coil 93 to spring arm tip 95. Tip 95 engages opening 79 of striker 70, preferably directly as shown or through another linking member in the immediate local position. Latch 50 is preferably pivotally attached to the tool assembly by recess 57 at handle hinge pin 110. In FIG. 3A, tab 54 of the latch 50 engages opening or edge 74 of the striker 70 to selectively immobilize the striker.

[0067] The motions of the parts described above are shown by comparing FIGS. 1 and 3. Pressing handle 20 about hinge pin 110 causes link 30 to move downward. Lever 40 pivots about hinge 41 to cause link bar 60 to move downward. The linkage assembly thus forces spring arm 92 to deflect downward or equivalent direction. Striker 70 cannot move downward from the action at latch 50, so spring arm 94 remains in an upper position as seen in the pressed position of FIG. 3. Power spring 90 becomes deflected with spring arm 92 spaced away from spring arm 94. Power spring 90 is thereby energized for an operating cycle to eject fasteners from track 180. In FIG. 3A, hinge pin 22 has just made contact with tab 53 of latch 50 while handle 20 is in a low but not lowest position. Moving the handle farther toward the lowest position of FIG. 4 starts the latch 50 rotating to disengage striker 70 as described next.

[0068] In FIGS. 4 and 4A, the pre-release condition has latch 50 disengaged from striker 70. Tab 54 is moved away from opening 74 so striker 70 is now free to move down. It is preferred that the release of striker 70 occur as close as possible to the handle's lowest position. This handle lowest position, FIG. 4, is defined by contact to bumper 25 of handle 20 against a surface of housing 10 or equivalent action. Thus, there is minimal jump or jerking of the handle 20 upon release for reduced operator fatigue. Further, the force of an operator's hand presses directly on the housing body 10 through bumper 25 to help hold down the tacker as it fires. To move the latch 50 as described, the pin or equivalent structure of hinge 22 presses tab 53 of the latch. The latch 50 rotates about hinge pin 110 to slide tab 54 out from the striker 70. The preferred latch motion is precise, reliable and repeatable since it is directly tied to a short portion of the handle motion; the latch begins to move only during a late part of the handle stroke so its release motion is relatively fast during the relevant handle motion. Specifically, the latch release motion occurs only between the pressed handle position of FIG. 3 and the pre-release position of FIG. 4, this being about 1/2" at the handle rear for the exemplary model shown. With all of the latch release motion concentrated near the end-of-stroke, any tolerance variation of the pre-release handle position will be confined to a pre-determined position within this small portion of the handle motion. The latch 50 operates about a common pivot to the handle 20 so there is no tolerance variation of intervening components; the latch and handle move in unison during release. There is also minimal net vertical force on hinge pin 110 since handle 20 and latch 50 pull oppositely on the pin. Therefore, the pin 110 can rotate with handle 20 about its mounting on housing 10 with little force and friction at the housing mounting. This unified motion reduces friction between the latch 50 and the pin 110 as demonstrated in a working model and through empirical testing.

[0069] In FIGS. 4A, 9, the exemplary embodiment tab 54 has a preferred acute angle of approximately 89 degrees relative to an imaginary radial line extending from hinge pin 110. An angle within approximately 2 to 5 degrees of 90

degrees can be suitable to hold the latch 50 stable on the striker with minimal force on the latch required to move the latch as described. Through empirical observation, with the exemplary angle of 89 degrees, in the release action, rotating latch 50 under load as between FIGS. 3 and 4, adds less than 1 lb., being about 1/2 lb., to a peak handle force. This force is effectively undetectable to a user. When measured at the handle rear in the position of FIG. 4, the required total force is approximately 15 to 16 lbs. to provide power sufficient to drive 1/2-inch T-50 type staples flush in common construction wood applications, for example, Douglas fir wood. Therefore, the tacker provides tremendous staple driving energy while the handle deflection effort as perceived by the user is very low and smooth.

[0070] The linkage between handle 20 and striker 70 is substantially rigid through the structures described here. In the spring rest condition of FIGS. 1, 14A and 14C, pivot/support element 96 presses spring arm 94 to hold power spring 90 preloaded. FIGS. 3 and 14B show the power spring deflected and energized. Pivot element 96 is preferably a laterally extending portion of a spring arm and may be referred to as a "location of preload" or of preload force for the spring, such location being spaced from coil 93 to enable a preload torque on the coil. The lateral direction is into the page in FIGS. 1 and 18, being preferably, but not necessarily, perpendicular to arm 94 in FIG. 14C. This spring arm crossing, FIG. 14C, may be at shallower angles. The pressing is preferably directly between the respective arms 92, 94 while the arms may also press in the local area through further elements. Pivot element 96, preferably but not necessarily along with tip 91, forms a hook to hold the spring in the preloaded condition at the location of preload. As the handle is pressed by the user, pivot element 96 is forced downward. The force on striker 70 at tip 95 increases from near zero to a final maximum at the pre-release position of FIG. 4. This force is a torque on spring arm 94. The spring arms 92, 94 are of a functionally and intentionally resilient material being normally of a same wire as the coil. However, flexing forward of the location of preload is not useful as discussed below; hence the length of the portion forward of pivot element 96 is minimized in the preferred embodiment.

[0071] To demonstrate this minimized forward portion length, in FIGS. 1, 14A to 14C, spring arm 94 flexes in proportion to the length of the unsupported cantilevered segment between pivot/support element 96 and the striker location at tip 95. This effect is illustrated in FIG. 14B: in phantom lines, support element 96 is pressed down slightly from FIG. 14A until element 96 no longer presses spring arm 94. Spring arm 94 flexes as shown until support element 96 is no longer in contact at S1. With the loss of support at S1, the support shifts farther forward to the striker at S2. This flexing to remove the preload translates to handle 20 as a mushy start to the stroke and lost energy input as discussed below relative to FIG. 35. It is thus desirable to have S1 be as close as practicable to S2 as shown and separately discussed to minimize the effect of this flexing.

[0072] As illustrated in FIG. 1, a distance between mandrel pin 107 or equivalently a central axis of the coil, or a spring coil central location, and striker 70 is approximately 2.06 to 2.11 inches. Most preferably, this is a distance of approximately 2.11 inches, and is denoted by dashed line L1 in FIG. 1. In this context, the striker location is defined as the rear plane of the blade of the striker at engaged opening 79. From support element 96 to striker 70 is a distance of

approximately 0.43 inch, as denoted by line L2 in FIG. 1. L3 is the distance between mandrel pin 107 and support element 96, and L3 in this embodiment is approximately 1.70 inch. There is a distance ratio $L3/L1$ of about 80% (i.e., 1.70 in./2.11 in.). Therefore, the location of preload is forward of the coil location about 80% of the length of dashed line L1. In FIG. 4 this distance places support element 96 adjacent to striker 70 in the pressed spring condition, preferably clear of sidewalls 72 or other striker structure by not more than one spring wire diameter, although other spacings to the striker are contemplated. A distance ratio $L3/L1$ of more than 50% is preferable, while a distance ratio of more than about 60 or 70%, is more preferable to have spring arm 92 terminate adjacent the striker and thereby see the benefits described below, based on empirical observation. Other dimensions are contemplated in proportion to other overall tool sizes. The foregoing ratios or proportions are with reference to the rest position of FIG. 1 although they are not substantially different in the released position of FIG. 5.

[0073] Flexing of cantilevered spring arm 94 as described above is felt as a “dead bounce” at the handle—a mushy feel that is minimized in the present invention as discussed above regarding FIG. 14B. Based on empirical observation and mechanical principles, this flex is a waste of handle travel and useable energy input as illustrated in the x-y plot of FIG. 35 discussed in further detail below. With such flex minimized, handle 20 is effectively rigidly linked to power spring 90 at a location just approximately 0.43 inch from striker 70 through the four-bar style cantilevered linkage, or an alternative linkage arrangement, discussed below. With the short cantilever L2 of the “beam” of spring arm 94 as described, there is minimal beam flexing and no perceived dead bounce. Therefore, user effort on the handle is perceptibly reduced, and the smooth operation of the handle greatly improves the feel of the tool for the user.

[0074] FIGS. 14A to 14C show various views of a preferred embodiment power spring 90. In FIGS. 14A and 14C power spring 90 is in a preloaded rest condition. Pivot/support element 96 is pressing spring arm 94 in proportion to a preload selected for the particular power spring characteristics. Accordingly, there is a free position (i.e., unflexed) for the spring wherein spring arm 92 is preferably angled upward and pivot element 96 is spaced above arm 94 relative to the view of FIG. 14A. A pre-assembly step has link bar 60 (FIGS. 1, 2, 12) assembled to power spring 90 with pivot element 96 passing through hole 66 in link bar 60. In the pre-assembly step, the spring arms are then forcibly moved from the free position to the position depicted in FIGS. 14A and 14C to form a sub-assembly of link bar 60 and power spring 90 wherein the spring is preloaded. Tip 91 of power spring 90 preferably passes beside spring arm 94 to secure spring arm 94 on pivot element 96 and to hold the assembly stable. The assembly preferably has tip 91, link bar 60, and spring arm 94 laterally adjacent each other along pivot element 96.

[0075] An alternative embodiment tool may use a power spring in the form of a single or assembly of flat bar springs instead of a coiled wire torsion spring. The bar spring includes cantilevered legs and is preloaded similar to FIGS. 14A to 14C. The bar spring is mounted to a mandrel 107 or like fixture inside the housing.

[0076] In the present embodiment tool, a “four bar” or equivalent rigid linkage forms the linkage assembly to connect the rigid steel handle or equivalent rigid structure to

pivot element 96 of power spring 90. In the four-bar assembly, lever 40 is pivotally mounted at its rear at hinge 41, depicted in FIG. 1. Link 30 presses lever 40 toward a central portion of lever 40 at hinge 43 and the lever presses link bar 60 at a front distal end of lever 40. Lever 40 is cantilevered forward from its links at hinges 41 and 43 and therefore lever 40 can extend forward to a striker proximate position. In this manner, a vertical linear motion from the handle at link hinge 22 may be enhanced at pivot tab 45, and thus on pivot element 96 or equivalent structure, through cantilevered lever 40. As shown between FIGS. 1 and 3, the vertical travel at link hinge 22 is about doubled at pivot tab 45 since the lever is pressed near its center. However, if hinge 41 were located farther rearward in housing 10, this doubled travel decreases, with a factor of 1.1 still allowing usable lever geometries. Spring pivot element 96 and lever pivot tab 45 are substantially vertically aligned so pivot element 96 maintains the preferable, at least 80% distance ratio discussed above. Thus, pivot element 96 is also proximate the striker as described.

[0077] All of the linking elements of the linkage assembly described here may be made of steel so there is no obvious or perceptible give or play in the system beyond that used to store spring energy. It is apparent from the above geometries that handle 20 should rigidly link to power spring 90 at a most forward position of the power spring. As shown in FIG. 1, this link at the location of preload adjacent to pivot element 96 is substantially aligned vertically with handle hinge 110, indicated by vertical line L in FIG. 1, whereby there is a position of line L that passes through or near tangent with both pivot element 96 and hinge 110. Described another way, line L is substantially vertically coincident with each of hinge 110 and pivot element 96 (preload location). Similar considerations apply to FIG. 16 for example. Similarly, link bar 60 extends vertically in or near alignment below handle hinge 110, being vertically coincident in this alignment as shown wherein a top view has some structures of hinge 110 overlapping structures of element 96.

[0078] In the four-bar system depicted in FIGS. 1, 2 and discussed above, there is a rear bar including the structure of housing 10 supporting spring mandrel 106 and hinge 41, a front bar in the form of link bar 60, a top bar being lever 40, and a bottom bar being spring arm 92. Link bar 60 is pivotally guided within this four-bar system by pivot element 96 of the power spring, FIG. 4B. The torsion spring as described is thus particularly suited for the present four bar system. Spring arm 92 provides both an interface to energize the spring and also a functionally rigid member of the four bar system to guide the lower end of link bar 60. These combined functions are not possible, for example, with a compression spring which is inherently unstable in lateral directions.

[0079] FIG. 35 is an x-y plot depicting empirical observations of unexpected results and benefits of the rigid structure described above. The plot shows comparative test results from working models of torsion spring tackers with similar tacking performance. It is based on measurements of force F at the distal or rear end of a handle (y axis) versus the distance D (x axis) that the handle moves, with initial handle free play omitted, but “dead bounce” included. The areas under the respective curves correspond to energy stored in the power spring. The “Long Arm” sample plot has a first spring arm pressed in preload by a second arm about halfway between the coil and the striker, an arrangement that

would have L2 and L3 of FIG. 1 being close in value. In contrast, the “Short Arm” sample plot has the ~80% ratio discussed above, being pressed in preload closer to the striker. A steep initial slope in the Short Arm plot indicates a stiff linkage with reduced dead bounce and a quick start to energy storage (as shown in phantom in FIG. 14B and discussed above). The shallower slope of the Long Arm plot shows extra flexing or give between the handle and the power spring. As seen, there is substantial wasted handle motion up to about 0.4 inches of travel for the Long Arm, so the Long Arm tacker requires a higher handle force for similar performance. Accordingly, the exemplary embodiment Short Arm tacker enjoys measurable performance advantages over Long Arm tacker designs.

[0080] The exemplary embodiments disclosed herein include a tensile link between the striker and the handle while enabling easy assembly of the tacker tool. This is further advantageous in that if the striker becomes stuck in a lower position, it is possible to forcibly move the striker upward by pulling the handle with a tensile force. As seen in FIG. 1, the link between link bar 60 and power spring 90 at hole 66 is inherently multi-directional. The next connection is between link bar 60 and lever 40. This connection is between pivot tab 45 and groove 65 of link bar 60. During assembly, lever 40 is rotated counterclockwise about this connection to engage tab 68 above catch 48. The tab and catch remain engageable for all operative positions—compare FIGS. 1 and 3A for example. There is a small clearance to tab 68 to ensure normal compressive operation has only pivot tab 45 and groove 65 engaged. When lever 40 is pulled upward, catch 48 presses tab 68 from below to pull link bar 60, and thus the power spring and striker, upward.

[0081] Re-set spring 190 biases the relevant moving parts toward the rest condition, FIGS. 1 and 2, in normal use. The re-set spring 190 pivots about leg 194 in hole 157 of absorber 150, as per FIGS. 1 and 15. In FIG. 2, absorber 150 is omitted to show underlying elements. In FIG. 4B, at its upper end, angled leg 193 engages opening 67 of link bar 60, with an angle of leg 193 biasing spring arm 192 to be retained in the opening.

[0082] The components including everything below link 30 are preferably initially assembled so that the lower tacker structure is complete, including both housing halves and front cover 12. Only parts associated with the handle remain to be attached so that there is no need to hold various lower parts in position as the handle is manipulated into the assembly. This eases assembly effort for volume production.

[0083] An upper subassembly includes handle 20, bumper 25, link support 28, latch bias spring 130, and link 30, as in FIGS. 1, 2. Latch bias spring 130 is supported about hinge pin 22 at spring coil 133 and held in position at rear end 134, as in FIG. 3A. These parts are pre-assembled to handle 20. Link 30 hangs loosely from handle 20 about link hinge 22 before installation to the lower tool structure. In FIG. 2, link hinge pin 22 naturally forms a multidirectional link within respective holes of the two connected parts. Pin 22 also supports latch bias spring 130 in this pre-assembly. As the handle sub-assembly is installed, the elements of the lower structure are in the rest condition of FIG. 1. Latch 50 is placed atop lever 40 to rest against angled face 75 of striker 70, in the approximate position shown in FIG. 1. The tacker body and handle are positioned with the tool front angled upward to allow a lower end of link 30 to drop over hinge 43 at slot 33 of the link. Hinge pin 43 (FIGS. 3A, 10) is a

pre-installed pin of lever 40. Rotating the link allows the handle to line up at hinge pin 110 at which point pin 110 is installed to support both latch 50 and handle 20. This process is demonstrated to be effective in a working model. In FIG. 3A, it is seen that rib 37 of the link now cooperates with lever tab 47 so that pulling up on handle 20 causes rib 37 to press tab 47 from below to transmit a jam-releasing tensile force. Therefore, the preferred embodiment tacker benefits from an anti-jam tensile force available to link striker 70 to handle 20 operated by the user. Optionally, some or all the functions of link support 28 may be integrated into a handle structure, for example, in association with a molded polymer composite handle. For example, recesses in sidewalls of the handle could support link hinge pin 22 with latch bias spring 130.

[0084] The staple is driven from the present invention tool, and now in a re-set action, striker 70 moves from its low released position of FIG. 5 to its upper rest position of FIG. 1. In FIG. 5A, it is seen that moving striker face 75 upward will cause latch 50 to rotate counterclockwise in the view. This cam action persists until latch tab 54 lines up with striker opening 74, such as FIG. 3A. Latch 50 then rotates clockwise under the bias of re-set spring 130 as the tab enters opening 74 to assume the position of FIG. 1. Latch 50 now selectively holds striker 70 in its upper position. Tabs 55 contact face 75 to hold latch 50 in a position in opening 74 that clears a radius at the base of tab 54, as in FIG. 1. In the conditions shown in FIGS. 5A and 6, striker 70 is down and out of engagement to latch 50. As handle 20 rises in the re-set stroke, the latch tends to rotate clockwise from re-set spring 130. In FIG. 3A, latch 50 has a stop against the housing formed by housing notch 11 against latch tab 56 to limit this rotation to the operative position shown in the drawing when the striker is not present. Thus, tab 54 of latch 50 remains in a position forward of face 75 whereby the re-set cam action of the latch and striker may occur.

[0085] In FIG. 5, striker 70 includes a blade or plane defined by its position at 78 immediately in front of track 180. It is preferable to minimize any element of the tool that extends forward past this position 78 to ensure a staple can be installed reasonably near a confining wall, corner, or like obstruction. Further, a compact front of the tool maintains a beneficial line of sight for the user for aiming the tool. In FIGS. 1, 5A and 13, the tool includes an optional hump 12b in front cover 12 to clear power spring arm tip 95. Also handle 20 extends forward in its pressed position, FIG. 5A, but no farther than hump 12b. To limit the handle or like extension, latch 50 engages striker 70 at a position behind blade 78 of striker 70. To do this as seen in FIG. 4A, striker 70 includes a dogleg or offset bend 76 whereby opening 74 is preferably spaced rear of blade 78 or main striker structure. Latch 50 then may rest and move rearward of the blade and/or cover 12, as in FIG. 1. Latch 50 is located near or at a top of striker 70 as shown in FIGS. 1, 2. Latch 50 being disposed in the upper location of the tool is clear of the area occupied by re-set spring 190, absorber 150, and tabs 71, as discussed below. By utilizing this arrangement, there is abundant space in the front lower area of the housing behind the striker for these further parts to be assembled, to operate, and to function well.

[0086] In FIG. 2, to provide an impact stop against absorber 150, striker 70 includes horizontal tabs 71 bent from side walls 72. These tabs 71 contact absorber 150 in the low striker position of FIG. 5 where striker end 78 is at a

bottom of the tacker body. In FIGS. 6, 11, to reinforce tabs 71, striker 70 includes extensions 72a in contact with blade 78 at 70a immediately above the tabs 71. These extensions 72a provide a direct force path from the moving bodies of striker 70 and power spring tip 95 to tabs 71 to reduce bending stresses on the blade structure where the tabs meet side walls 72.

[0087] It is common in a torsion spring tacker design that an absorber acts directly on an arm of the power spring—in particular, that a dry fire, without staples, has the absorber directly stopping an arm of the spring rather than the striker. This causes an undesirable reversal of forces in the spring arm type absorber. In normal use when the tool is fired, spring arm tip 95 presses down at the striker hole 79 (FIGS. 6, 11) to install staples. But with the spring arm to absorber contact structure in a dry fire there is a reversal of force at the spring arm/striker interface. The spring arms stops first, and the striker overshoots the spring arm a small distance at hole 79 and impacts the spring arm at the top of the hole to be indirectly stopped by the absorber.

[0088] This overtravel action causes wear at hole 79 from both top and bottom, leading to a stretched, distorted, or enlarged hole, added tensile stress on the striker, and increased vertical free play of the striker about the spring arm. In the extreme, the hole is so ovoid that spring arm will not be able to raise striker high enough to set the latch or reach a release height. As described herein, absorber 150 acts directly on striker 70. Therefore, striker 70 is always one of accelerating, pressing a staple, or pressing the absorber. Spring arm 94 at tip 95 thereby always presses down within hole 79 and thus wears the hole in only one direction with minimal tensile stress on the striker in this area. From empirical observations, this arrangement improves longevity and operating life of the tool.

[0089] Further, in the case of a spring wire/absorber interface, the wire spring arm provides a small impact target for the absorber leading to high stress in that contact area. In the present preferred embodiment, any target area on power spring arm 94 is further interrupted by the useful forward location of pivot element 96 at the front distal end of spring arm 92, corresponding to a short L2 length in FIG. 1. While keeping this segment L2 short, which is useful as discussed, it provides a small absorber target. With the absorber contact being against a structure of or affixed to the striker, the absorber can directly vertically underlie the distal end of arm 92, for example, at pivot element 96. As best seen in FIGS. 1 and 5, absorber 150 extends rearward of pivot element 96. This structure may be described as having an alignment along a vertical line of at least absorber 150 and the distal end of arm 92 with handle hinge 110 also preferably so aligned above absorber 150, and with spring coil 93 rearward of this alignment. In an alternative embodiment, there may be an additional or only absorber contacting arm 94 or other structures that move with the striker.

[0090] As shown in FIG. 11, the impact stop (horizontal tabs 71) is bent directly from the material of the striker 70 to preferably minimize weight and inertia of the reciprocating impact parts, although separate components may be used. It is desirable that the mass of the striker and any other parts that move in the impact or firing stroke be minimized. When these parts are kept light weight, the tacker installs staples and the like more effectively, especially when the tacker is actuated with a single hand. Consequently, the body including housing 10 will not jump substantially upward as

the staple exits since the body is very heavy compared to the fast moving but light weight striker. This gives the user a damped, less jarring feel from the tool with reduced hand fatigue. As shown in FIG. 11, striker 70 includes optional openings above and below spring opening 79 to further reduce its weight.

[0091] Housing 10 preferably includes two halves. A left half is shown in the views of FIGS. 1 to 6. The halves must be secured in a properly spaced relation for effective tool function. In FIGS. 1 and 2, mandrel 106 is supported by pin 107. This pin 107 may be a screw or rivet to compress the housing about the mandrel. Mandrel 106 thus holds the housing securely spaced apart for operating clearance for spring 90 and further holds the housing halves from sliding relative to each other. At the lower front of the housing in FIGS. 1 and 2, plate 155 holds the housings apart while front cover 12 clamps the housing from in front. Housing plate 155 preferably supports rubber absorber 150 in an absorber assembly, shown in FIG. 15. In the cut away cross-section of FIG. 1, track chamber tab 129 extends within slot 156 of plate 155. FIG. 2 also shows these parts with absorber 150 omitted for clarity. Tab 129 in FIG. 1 provides an accurate rear limit position relative to track chamber 120 for striker 70 in its upper rest position. In FIG. 2, striker 70 is laterally positioned by edges 158 of plate 155. To register plate 155 to track chamber 120 laterally, tab 129 is a close fit in notch 156. Thus, there is essentially no tolerance build up from a more indirect link of the plate to the striker and track through the housing enclosure.

[0092] At the front top there is minimal room for a similar plate since, for example, latch 50 is advantageously located there. Preferably, as seen in FIGS. 6 and 13, front cover 12 includes registration notches 19 to cooperate with housing tabs 17 during assembly. With tabs 17 secure in notches 19, the housing is held accurately spaced part in this area.

[0093] In the drawings and disclosure, a single power spring is shown. In alternative embodiments, there may be two or more of such springs. For example, two coiled power springs 90 may be vertically stacked with a second mandrel 106 below the first mandrel, in front of grip opening 18 in housing 10. Pivot element 96 of this second spring engages a second link bar hole 66 (not shown) below the first. In this alternative embodiment, the horizontal distance between mandrel pin 107 and hole 66 (for both springs) is close to the same as that between hinge 41 and pivot tab 45. This ensures that pivot tab 45 and both holes 66 remain aligned through their motions to prevent binding. In another alternative embodiment, two power springs 90 may be installed side by side axially on a common mandrel 106. As with the other disclosed embodiments, power spring 90 is pivotally attached to the housing near or forward of a front of grip opening 18 whereby arms 92 and 94 form torque arms and extend from this position to striker 70. With relatively short torque arms, there is high force available at striker 70 for useful work, and further there is minimal vibration in the arm action as the short arms operate. If desired, longer arms may be used, with a more rearward mounting. Arm 94 may be described as a first spring arm while arm 92 may be described as a second spring arm.

[0094] In FIGS. 1A and 13, front cover 12 includes raised bottom front edge 12a. This raised portion may extend along the side walls of cover 12 rearward across striker slot 13. In use, a tacker is often held at an angle to the work with the rear end held up. With the clearance described here, the

striker end **78** (FIG. **5**) can still extend close to the work piece with front cover **12** out of the way. This front edge **12a** may be raised by approximately 0.020 inch for example. With the light weight reciprocating compact parts discussed above and the tight contact here, ordinary stapling will easily produce driven staples that are flush with the work surface. Workpieces having such fully installed staples will hold the work more tightly and have a higher quality workmanship.

[0095] FIGS. **16** to **23** show a second exemplary embodiment of the present invention. Many elements may be shared with the first embodiment described above and the mechanical actions of power spring **90**, striker **70** and latch **50a** are or may be equivalent. Distance ratios described in connection with the first embodiment may be employed in this second embodiment as well. Also, the geometries that the handle should rigidly link to the power spring at a most forward position, proximate vertical line **L**, may be applied in this embodiment. Finally, the part count, friction, and complexity are or may be reduced in the second exemplary embodiment.

[0096] In FIG. **16**, handle **20** to lever link **330** pivotally connects handle link support **328** to lever **340**. Lever **340** directly engages pivot element **96** of power spring **90** at opening **366**. Opening **366** may be elongated to provide for longitudinal (left-right on the page) motion of power spring **90** relative to lever **340** at this location. Lever pivot **341** operates rearward of spring coil **93** while lever **340** extends along a lever length forward past spring coil **93** to be disposed adjacent to striker **70**. Link **330**, at hinge **333**, presses lever **340** at central lever pivot **343** toward a central location of the length of lever **340**. Lever **340** is thus cantilevered forward from central lever pivot **343** to the spring location of preload on pivot element **96**. In this manner, opening **366** is proximate the location of preload, being laterally adjacent (into the page of FIG. **16**) along pivot element **96**. At least one of spring arms **92** and **94** are likewise cantilevered forward from spring coil **93** whereby each of power spring **90** and lever **340** are cantilevered forward to the location of preload. As shown in FIGS. **16-18**, both spring arms **92**, **94** are so cantilevered.

[0097] The second exemplary embodiment depicted in FIGS. **16** to **23** may provide further reduced friction and increased rigidity over that of the first exemplary embodiment in FIGS. **1** to **6**. While the first embodiment is substantially rigid, as seen in FIG. **35**, the second embodiment has one less component between handle **20** and power spring **90**, and thereby fewer pivotal or other connections to introduce flex or free play motions. Lever **340** is also longer than lever **40** and therefore rotates through a smaller angle about its rear pivot **341** to move power spring **90**. From empirical observations, it is about 12 degrees for the second embodiment versus 20 degrees of pivoting for the first embodiment. With less motion there is less friction at the hinge of the rear pivot.

[0098] A similar effect operates when comparing the central pivots of **43** and **343**, respectively. In FIG. **18**, lever front opening **366** rotates in a same direction as spring pivot element **96** to reduce sliding there between, thereby reducing friction over the structure of FIG. **1** where link bar **60** does not substantially pivot along with pivot element **96**. Whether considering the second embodiment, FIGS. **16** to **23**, or the first embodiment of FIGS. **1** to **6**, each provides substantial improvements and benefits in function and utility over the prior art, for example, through a rigid linking

system as disclosed. According to this rigid linkage system, the lever front end presses the second arm at a lengthwise position substantially closer to the striker than to the spring mandrel center. This pressing occurs at a front end of lever **40**, seen in FIG. **1**, or lever **340**, seen in FIG. **16**.

[0099] The second embodiment of FIGS. **16** to **23** further enjoys simplified assembly. In FIGS. **19** and **23**, link **330** is installed at hem **332** or equivalent structure into slots **329** of handle link support **328** while the parts are loose as depicted in FIGS. **20** and **21**. Link support **328** is then fastened to handle **20** by riveting or the like, as in FIG. **16**. Link **330** is thereby pivotally confined on handle **20**. Hem or top end **332** presses and pivots against an underside of the handle as in FIG. **17**. This pivoting is minimal, about 4 degrees as shown, so friction is low and slot **329** can be narrow. With the upper and lower respective assemblies prepared, seen in FIG. **18**, the handle assembly is lowered into position as shown. Link **330** is held about at the angle shown to align with the front wall of notch **344**. Link **330** is held out of the page in FIG. **18**, and/or lever **340** pressed in, so that the link may pass beside lever **340** to assume the position of FIG. **19**. In both FIGS. **18** and **19**, handle **20** to body pivot **27** is forward of its final position. In FIGS. **21** and **22**, tab **335** of link **330** can be seen able to enter notch **344** of lever **340**. As seen in FIG. **19**, handle **20** is then moved rearward to its final position at pivot **27** as link **330** rotates to be guided by edge **348**. Edge **348** then locks link **330** laterally, in a notch of the link at tab **335**, to a pivoting relation on the lever with respect to the side views. Link **330** holds lever **340** stable laterally through a triangular geometry “T”, as seen in FIG. **23**. Hem **332** presses inside handle **20** to form a stable base of the triangle. The pivoting is at link hinge **333** against central lever pivot **343**, as seen comparing FIGS. **16** and **17**. A tensile link between handle **20** and power spring **90** operates through link **330** at slot **329** and edge **348** to enable the handle to pull up on the spring and striker in an event of a staple jam or the like.

[0100] Spring arm tip **95** is preferably on center with respect to a front view to press striker **70** at its center line, although an off-center alignment can also be functional. Lever **340** therefore presses spring element **96** off center at pivot **366** in a similar position as link bar **60**; see FIG. **4B** for this analogous position at **66** in the first embodiment. Lever **340** is therefore preferably off center, into the page in FIG. **16**, at its three operative pivots **341**, **343**, and **366** to form a stable plane of action. Segment **349** may be on center, out of the page in FIG. **18**, to keep its optionally exposed portion at a clean joint line of housing **10**.

[0101] In FIG. **16**, latch **50a** operates similarly to latch **50** as disclosed with the first embodiment. Rear end **53a** selectively contacts handle **20** to cause the release action. In FIG. **19**, link tab **322** supports latch bias spring **130** at coil **133**.

[0102] FIGS. **24** to **34** show an exemplary embodiment staple guide track and loading system preferably used with the first and second embodiment tackers described above while also providing advantage for use with other tacker devices. The subassembly shown provides for bottom loading staples or other fasteners, as in FIGS. **31-33**. As seen in FIG. **33**, track **180** selectively extends rearward to expose staple holding channel **128**. The track can extend farther wherein track guide tab **188** contacts stop rib **125** of track chamber **120** or equivalent structure. Preferably, the full extension has tabs **188** at least about 4 inches rearward of front cover **12** to fit a standard staple rack **405** of that length.

In FIG. 33, staple rack 405 is shown in position to be placed in track chamber 120. Rack 405 is shown as about half standard length corresponding to the partially extended track shown.

[0103] This exemplary embodiment bottom loading system is advantageous over a rear staple insertion system, because bottom loading keeps any staple readily accessible when needed. For example, it is much simpler to clear a staple jam or malfunction because, as seen in FIG. 33, the staple channel may be exposed for easy manipulation or extraction of such staples. In contrast, a rear loading system requires dismantling the track subassembly to access any jammed staples at the front of the tool.

[0104] The structure of the present track subassembly is suited for use with a sheet metal bodied tacker, although it is not limited to that application. For example, it may be used with die cast or molded bodied tackers. The preferred embodiment track subassembly includes closely integrated track pull 160 which de-latches track 180 from its operative position of FIGS. 24 and 31, for example, to its de-latched position of FIG. 25 through a simple pull rearward. Grasping and pulling track pull 160 (FIG. 27) causes it to rotate about pivot 161 against a bias from latch spring 140, discussed below, to the position of FIGS. 25 and 32A. Continuing the same pulling action causes track 180 to move to the extended position of FIG. 33 while the track pull preferably returns to its normal upright or equivalent position under the latch spring bias. Pushing track pull 160 inward, to the right in the views, moves track 180 to its closed, operative position as in FIG. 31. The track becomes latched to be retained in position while the track pull remains upright or otherwise in its normal position relative to the track through a latching action.

[0105] The views of FIGS. 24 and 25 have track 180 and track chamber 120 shown in lengthwise cross-section to expose the internal workings. The latched track condition is seen in FIGS. 24 and 24A. Rib 184 of track 180 (see also FIG. 30) engages detent 124 of track chamber 120 or equivalent structure (housing 10, for example). In FIGS. 24 and 28, spring front end 146 is held on at track support 181 and held centrally at spring loop 143 by fulcrum 186. See also FIG. 33 for fulcrum 186. Latch spring 140 is therefore cantilevered at rear end 147. Cantilevered spring rear end 147 presses downward (upward in the page of the inverted views of FIGS. 31-33) on track spring contact tab 126, FIG. 24. This is beneficial because track 180 is thus resiliently urged upward relative to the track chamber to press track rib 184 against detent 124. Upward in this context means toward the handle from the track area for any view orientation. Preferably, track pull arm 167 also contacts spring end 147 in the fully closed track condition so that the track pull does not rattle.

[0106] Track pull 160 is pulled rearward to open track 180 to the position of FIGS. 33, 34. It is natural to squeeze and pull it at sides 166 or pull the front edge in this area, near part number “166” in FIG. 24A. Track pull 160 rotates about hinge 161 to the position of FIGS. 25, 25A. In FIG. 25, arm 167 has deflected spring 140 upward at end 147. Pulling outward on track 180, indicated by an arrow in FIG. 25, causes a downward bias on the track by a cam action from the angle of detent 124 and rib 184. Spring 140 does not resist this down motion since the spring is deflected off of

tab 126 by arm 167. As a result, track 180 clears detent 124 as shown and is free to slide rearward to the position of FIG. 26.

[0107] It is not required that the track pull rotate to deflect the latch spring. Optionally, the track can be directly pulled down to deflect the spring and clear detent 124 before pulling out, for example, through a track pull interface that cannot rotate. While this optional structure does function, it requires two steps. In contrast, the preferred track pull 160 provides an automatic cam action that provides the down motion automatically through a single step of an intuitive outward pull. These features have been demonstrated in a working model.

[0108] As best seen in FIG. 32A, track pull 160 is positioned laterally by arm edges 167a within track wall portions 185. The track pull is preferably a closely integrated fit to the housing body as seen in FIGS. 1 and 31. The tool maintains a clean profile in the rear area, being absent any track release access cavity, for example. Track pull 160 may include or comprise sheet metal, die cast, plastic molded construction, or any combination thereof. In any embodiment, the staple track remains easy to operate by a simple pulling action as discussed.

[0109] As track 180 is closed, following the arrows in FIG. 25B, latch spring 140 is deflected by the cam action at detent 124 and rib 184b causing the track to move down toward tab 126. The tab deflects the spring and moves the spring away from arm 167. In this manner track pull 160 remains, or at least may remain, in its upright position as an operator pushes the track pull in a normal manner. If, for example, the track pull required rotating outward during this motion it would oppose the operator's inward pushing force and would tend to lock up the system. Instead, the track pull remains stable, the action intuitive, and the closing operation ends in a satisfying and positive click.

[0110] Latch spring 140 in FIG. 28 may be a simple wire form as shown. As discussed above, front spring end 146 rests on track support 181. Notch 186 in FIG. 33 forms a fulcrum to hold spring loops 143 whereby spring 140 is preferably preloaded in its rest condition of FIG. 24 to keep the track pull rattle free and to hold the track securely in the closed position. As further seen in FIGS. 24, 25, 25B, latch spring 140 is slightly bent concave upward from the preload so that it is in contact or near contact with both arm 167 and tab 126. Pusher spring 200 biases staple rack pusher 400 toward the front of the track, thus urging the staple rack toward the striker.

[0111] Pusher spring 200 is attached in a known manner to pusher 400. The rear end of pusher spring 200 is preferably fitted to latch spring 140 at loop 202 as shown in FIG. 25. To install the latch spring to the track, track spring 140 is inserted to loop 202 and then guided by grasping pusher spring 200. Latch spring 140 is pressed into the channel of track 180 to deflect the cantilevered arms of spring 140 toward each other. When loops 143 are aligned with notches 186, the latch spring snaps into position. This has been demonstrated in a working model. A pulley at recess 189 (FIG. 30) may guide the pusher spring at front. Recess 189 forms an upward facing edge to support a shaft of the pulley in the track channel. In this manner, the pulley can be installed from top into the channel to rest on the edges of recess 189 rather than being installed from a side.

[0112] In track 180, notches 182 provide clearance for stop ribs 125 as the track is deflected downward, this being

up on the page in the inverted tool view of FIG. 32. As seen in FIG. 32, stop ribs 125 have entered notches 182. Similar clearance is created at notches 183 (FIG. 32A) to clear spring contact tab 126 as the track pull is deflected. Notches 182 and 183 preferably include a ramp at the front as shown so that ribs 125 and tabs 126 are guided out of the notch as the track moves outward.

[0113] In a front most position track foot 187 contacts stop edge 123a, as in FIGS. 29 to 32. Preferably, this contact is configured to hold pressure at the cam contact area of rib 184 and detent 124, i.e., the rear cam feature presses foot 187 against edge 123a. Track chamber tab 127a engages an opening of front cover 12 to hold a position of the track chamber at front, FIG. 127a. At rear the chamber is held to the housing by a fastener in hole 127. Side channels 122 of the track chamber guide track feet 187. At the rear end of the track, tabs 187a preferably fold across the track and may be spot welded or the like to reinforce the track structure. Ribs 125 and tabs 126 are formed as part of track chamber 120. The features of the track chamber may instead be formed from a structure of housing 10, for example, a sheet metal tab of the housing and the like for a steel housing.

[0114] Staples are normally and properly installed into bottom-positioned staple channel 128, as in FIG. 33. However, it is possible that an operator may attempt to load the staples from top onto the exposed track of FIG. 34. In particular, if the staples are able to enter the housing or tool interior on the track from here, an operator may reasonably assume it is supposed to function this way. Of course, it cannot, as seen in FIG. 26; the staples would be rearward of pusher 400 with no way to reach the front of the track for use. Negative user reviews of products that have this defect affirm this issue.

[0115] To address improper staple loading, as seen in FIG. 34, there is an optional staple blocker 16 that protrudes into the channel of track 180. It is an element of housing 10 although other structures are contemplated. Also, it is both physically and visually clear to the user that installing staples from the rear is impossible, and the reason is readily seen. Installing this way is clearly improper, informing the user that the staples “go somewhere else” upon which the bottom staple channel is readily discovered. A blocking tab of the housing, or track chamber, may extend inward from a side to abut the outer side face of track 180 in this area. This is seen as tab 16a in FIG. 34. It is preferable that a blocker be visible at the tool exterior so that there is no ambiguity to the staple exclusion. Further, FIG. 34 shows optional track tab 184a. Track tab 184a extends outward whereby a staple rack cannot fit upon or past it. In the case of a full 4-inch staple rack, tab 184a makes it obviously impossible to place staples on the track from this direction. A shorter rack such as 2 inches may fit in the track portion in front of tabs 184a, but with tabs 184a and 16 collectively making placement here clearly impractical, the message to the user to look elsewhere is reinforced.

[0116] FIGS. 36 to 40 show another alternative embodiment of the high efficiency torsion spring tacker.

[0117] Power spring 90a, in its operative configuration, is elongated to extend from an engagement with striker 70 rearward to coil 90c. Coil 90c is located with a coil axis above grip opening 18 of housing 210 and behind a forward extent of grip opening 18. Grip body 211 of housing 210 includes the portion of housing 210 overlying grip opening 18 whereby the grip body is elongated front to rear below the

handle. Accordingly, grip body 211 fully surrounds coil 90c, FIG. 36A, confined atop by ceiling 211a and bottom by floor 18a.

[0118] As described, power spring 90a is elongated to be near to half a length of housing 210. A relatively long power spring 90a has an advantage of a large arc motion radius at spring tip 95, labeled “Arc” in FIG. 38, showing the operative motion of tip 95. This Arc has minimal lengthwise component (left-right in the FIG. 38 view) so there is minimal lengthwise sliding of the spring wire in its engagement to striker 70. Sliding here adds friction to the operating action. Another advantage of a relatively long power spring 90a is minimal angular deflection of the spring arms 92, 94 for a given vertical striker motion. Compare the angles of second spring arm 92 in FIG. 36 versus FIG. 38. Less deflection provides less stress on the spring wire, other variables being constant. Less stress improves durability and fatigue life of the power spring and stapler.

[0119] To provide a low stressed spring and high energy storage, power or torsion spring 90a should be made from good quality metal wire such as music wire and equivalents. In structure, the power spring is preferably made of an optimal combination of wire diameter, coil diameter, and coil quantity. A constraint is the power spring must fit within the height and width of grip body 211, an external width being W in FIG. 36A. The grip body height is constrained by a minimum grip distance, G in FIG. 38. In practice the geometries shown in FIG. 38 provide G preferably at ~44 mm and the grip body height at spring coil 90c is preferably about 27 to 29 mm inclusive of the limits. In a steel-bodied tacker, W is customarily less than about 1 inch, preferably about 0.9 or 23 mm in the illustrated embodiment. Other dimensions among these are contemplated, within for example, 20%. To fit in this profile, a compact spring assembly is indicated. Tackers may also be constructed of plastic such as nylon. In a plastic bodied tacker, the outside dimensions will be larger to allow for the thicker walls required. Beneficially, a plastic construction allows for local thin areas to accommodate local structures, but with limitations. So a compact internal design is even more useful in a plastic body.

[0120] Within the preceding constraints, empirical testing has shown specific structures are uniquely effective. For example, a spring of about 4 mm wire, a mean coil diameter of about 17-18 mm and a free position coil count of about 3-1/2 provide a durable and powerful energy storage system to fit the size constraints discussed above. A preferred power/torsion spring 90a according to these proportions is shown in FIG. 39. In FIG. 40, the rest condition of the power spring 90a has a coil count of about 3-1/2. A coil count of greater than 3-1/2 such as 3-3/4 or 4 or more are contemplated. As seen in FIG. 37, four adjacent coil windings are arranged, referenced as coil windings (a) to (d). Using the spring rear view as depicted in FIG. 37 for reference, power spring 90a may be called a four-coil spring wherein preferably four adjacent coil winding segments abut each other at a rear of the spring coil 90c.

[0121] In contrast to a three-coil spring, for example, four coil power spring 90a offers more energy storage and a flatter force profile. Since each coil has less deflection for a given arm deflection, a force-deflection curve according to the plot in FIG. 35 will be flatter to provide a lower spring rate and thus lower maximum handle pressing force requirement. This is compared schematically to the plot in FIG. 41

where the three-coil spring has a higher spring rate (slope) and max force than the four-coil spring.

[0122] The exemplary embodiment-fastening tool is a spring-powered staple gun (tacker) suited to install T-50 format staples. In one preferred embodiment, a tacker should be able to fully seat a 1/2" T-50 type staple into structural softwood. Power spring 90a in the tool of the present preferred embodiment is well suited to this performance. But a further consideration in power spring 90a is its durability. It follows that a further benefit of reduced peak force is improved spring cycle life. Small differences in maximum force can cause large differences in cycle life. To illustrate, the plot of FIG. 42 is an exemplary schematic S/N (stress/cycle number) curve. The N scale is typically logarithmic so the life effect of reducing the force is great. Thus, reducing the max force has a non-linear benefit in spring life. It is thus advantageous to include a maximum spring coil count within other constraints of a tacker spring and its environment as disclosed above.

[0123] It is empirically known that a three-coil spring that fits within a common steel housing grip body and provides the performance described here has a life of about 15,000 cycles. However, a life of at least 25,000 is a common minimum standard to reflect real world usage. Within the confines of grip body 211 as described above, a four-coil power spring provides the required durability while a three coil does not. As noted, a further benefit has a lower peak handle force as shown in FIG. 41. Thus, the preferred four-coil power spring as described has unique advantages over any other coil count.

[0124] In a preferred embodiment, as best seen in FIG. 36, lever 440 extends forward from mandrel pin 107 to a termination near striker 70. Opening 367 engages a re-set spring arm (of re-set spring 190 of FIG. 16). In this instance, mandrel pin 107, through locating mandrel 106, serves as a pivotal mounting for both power spring coil 90c and lever 440 whereby both a lever pivot axis and the spring coil axis are coincident or nearly coincident, subject to operational tolerances, between mandrel 106 and an inner diameter of coil 90c. Lever 440 thus need not pass above or below the power spring coil 90c, so the lever with coil assembly can fit within the confines of grip body 211. Preferably, spring coil 90c spans near the full width of the interior of the grip body 211, as seen in FIG. 36A. This interior width is preferably as small as about 20 mm or 0.8 inch. In FIG. 36A, spring coil winding (a) is adjacent to the interior wall. Opposed coil winding (d) terminates at coil end 93d, FIG. 40, wherein the wire of the spring transitions to first spring arm 94.

[0125] For a compact arrangement, the rear end of lever 440 is at least partly nested within the confines of coil 90c. In FIG. 36, an imaginary cylinder is shown in dashed line. Lever top edge 446 and bottom edge 447 define an upper and lower extent of the lever rear end. The rear most lever end connecting these edges is an arcuate edge segment and substantially concentric to mandrel pin 107. The three edges form a rear edge perimeter, which lies generally behind cross-section line 36A in FIG. 36. Thus, the lever rear end is in majority confined within an inner diameter of coil 90c, directly above and below the coil axis with this confining limitation defined by imaginary cylinder Cy. The lever rear end thereby is partly surrounded radially on its rear edges by the spring coil 90c, and more specifically imaginary cylinder Cy. A linking lever segment of lever 440 extends forward

beyond the coil inner diameter to engage link 330 at opening 343. FIGS. 36A and 37 show the nested lever end. In FIG. 36A, coil end 93d lies partly under lever bottom edge 447. Lever top edge 446 is similarly partly under the upper winding portion 93dd of coil winding (d). This structure is seen in FIG. 37 from a rear view, the lever rear end is partly hidden by coil (d).

[0126] Terminal coil winding (a) at one end of coil 90c includes a top coil termination at 93a while opposed terminal coil winding (d) includes a bottom coil termination at 93d that becomes a base for second spring arm 92. Terminal coil winding (d) further includes an upper winding portion 93dd. Upper winding portion 93dd is spaced from the sidewall of grip portion 211 while opposed top coil termination 93a is adjacent to and possibly contacting the opposed wall, FIGS. 36A and 37. More generally, upper winding portion 93dd is spaced farther from a nearest wall than is top coil termination 93a. The space beside upper winding portion 93dd allows room for lever upper edge 446 to pass beside the spring coil, FIG. 36. Forcing lever 440 downward deflects second spring arm 92 to store energy in the power spring 90a, FIG. 38. Additional clearance at spacer 107a provides space for lateral motions of the power spring 90a, shown in FIG. 36A, as it is deflected and pivoted; this motion is enabled by the concentrically nested lever arrangement described here. In FIG. 38, handle 220 includes a lower edge adjacent to a terminal coil winding, while the handle remains external to grip body 211.

[0127] As seen in FIGS. 36 and 36A, within the confines of grip body 211, there are preferably no moving components beyond an outer diameter of spring coil 90c in a vertical direction. Further, as described above, the vertical extent of the lever 440 at the coil axis is preferably confined within imaginary cylinder Cy. This structure is in contrast with the FIG. 16 embodiment, where lever 340 is positioned vertically above coil 93, passing entirely above the coil of power spring 90. In the present embodiment having a spring coil 90c contained within the vertical confines of grip body 211, there is no available space for a lever or other substantial movable component to extend to be above or below the spring coil, while the nested structure of lever and coil shown fits within the grip body. Described another way, the grip body interior is unoccupied by internal moving elements above and below the spring coil.

[0128] Accordingly, there is very little wasted open space inside the housing body, which translates to a very compact, hand-grippable tool. As described above with regard to FIGS. 36 to 42, a compact power spring and lever assembly provides high performance and durability while preserving versatile options for a shape and size of a grip body of a tacker fastening tool.

[0129] While the particular forms of the invention have been illustrated and described, it will be apparent that various modifications can be made without departing from the spirit and scope of the invention. It is contemplated that elements from one embodiment may be combined or substituted with elements from another embodiment.

1. A fastening tool, comprising:

- a housing with a top, bottom and sides, the housing extending longitudinally between a front and a rear;
- a handle pivotally attached to the housing at a handle/housing pivot, the handle/housing pivot being at an upper front location of the housing;

a fastener guide track disposed along the bottom of the housing;

a striker disposed at the front of the housing including an upper striker position above the track and a lower striker position in front of the track;

a grip body of the housing elongated front to rear below the handle;

a grip opening of the housing elongated front to rear below the grip body;

a power spring supported within the housing, the power spring being a torsion type including a spring coil, the spring coil being confined within the grip body;

the power spring having a first spring arm extending forward from the spring coil to a first spring end, the first spring end linked to the striker to move with the striker, a second spring arm extending forward from the spring coil to a second spring end, the second spring arm pressing the first spring arm at a location of preload to hold the spring in a preloaded condition, the location of preload being spaced forward of the spring coil to be adjacent to the striker;

a lever extending longitudinally from a lever rear end to a lever front end, the lever pivotally attached to the housing at a lever pivot near the lever rear end, the lever rotating about the lever pivot to move vertically at the lever front end within the housing including an upper lever front end position and a lower lever front end position;

the lever including a central lever location between the lever front and rear ends, the lever central location linked to the handle at a location of the handle rearward of the handle/housing pivot, the lever front end being cantilevered forward from the central lever location whereby pressing the handle downward causes the lever front end to move downward, the lever front end including a linkage to the second spring arm forward of the central location wherein the lever, at the lever front end, moves downward to force the second spring arm to move downward away from the first spring arm in the lever lower position; and

the lever rear end including an upper edge, a lower edge, and an arcuate rear most edge connecting the upper and lower edges to form a rear edge perimeter, the rear edge perimeter being at least partly nested within a winding of the spring coil wherein the winding at least partly surrounds the rear edge perimeter.

2. The fastening tool of claim 1, wherein a rear of the spring coil includes four adjacent coil winding segments abutting each other.

3. The fastening tool of claim 2 where a lever pivot axis and a spring coil axis are substantially coincident.

4. The fastening tool of claim 1, wherein a spring coil axis is located above the grip opening and behind a forward extent of the grip opening.

5. The fastening tool of claim 2, wherein an external width of the grip body at a location of the spring coil is less than about one inch.

6. The fastening tool of claim 3, wherein an external width of the grip body at a location of the spring coil is less than about one inch.

7. A fastening tool, comprising:

a housing with a top, bottom and sides, the housing extending longitudinally between a front and a rear;

a fastener guide track disposed along the bottom of the housing;

a striker disposed at the front of the housing including an upper striker position above the track and a lower striker position in front of the track;

a grip body of the housing elongated front to rear below the handle;

a grip opening of the housing elongated front to rear below the grip body;

a power spring supported within the housing, the power spring being a torsion type including a spring coil with a spring coil axis, the spring coil axis extending across a width of the grip body, the spring coil being confined within the grip body;

the power spring having a first spring arm extending forward from the spring coil to a first spring end, the first spring end linked to the striker to move with the striker, a second spring arm extending forward from the spring coil to a second spring end;

a lever pivotally attached to the housing, the lever linked to the power spring wherein moving the lever causes the arms of the power spring to deflect and store energy;

a rear of the spring coil includes four adjacent coil winding segments abutting each other; and

the grip body having locations within the grip body vertically above and below the spring coil axis that are unoccupied by moving components.

8. The fastening tool of claim 7, wherein the grip body includes sidewalls, and the spring coil extends across the grip body between the sidewalls from a first end of the spring coil to an opposed second end of the spring coil.

9. The fastening tool of claim 7, wherein the lever is pivotally attached to the housing at a lever pivot axis, and the lever pivot axis is substantially coincident with the spring coil axis.

10. The fastening tool of claim 9, wherein, directly above and below the spring coil axis, a lever rear end is in majority confined within an imaginary cylinder described by an inner diameter of the spring coil.

11. The fastening tool of claim 7, wherein the spring is elongated front to rear within the grip body.

12. The fastening tool of claim 11, wherein the spring coil is located with the spring coil axis above the grip opening and the spring coil axis is behind a forward extent of the grip opening.

13. The fastening tool of claim 12, wherein a lever rear end is confined within the grip body.

14. The fastening tool of claim 9, wherein a lever rear end is at least partly nested within confines of a spring coil winding.

15. The fastening tool of claim 7, wherein a first end of the spring coil includes a bottom coil termination, a wire of the spring transitions to a first spring arm forward of the bottom coil termination, a second end of the spring coil includes a top coil termination, a wire of the spring transitions to a second spring arm forward of the top coil termination, a terminal coil winding includes the bottom coil termination and an upper winding portion above the bottom coil termination, and the upper winding portion is spaced farther from a nearest sidewall of the grip body than the top coil termination is spaced from an opposed sidewall of the grip body.

16. The fastening tool of claim 15, wherein a lever is pivotally attached to the grip body at a lever pivot axis, the

lever pivot axis is substantially coincident with the spring coil axis, the lever, including a lever top edge, extends forward from the lever pivot axis, and the upper winding portion is spaced apart from the nearest sidewall of the grip body to enable the lever top edge to pass through a space between the nearest sidewall of the grip body and the upper winding portion.

17. A fastening tool, comprising:

- a housing with a top, bottom and sides, the housing extending longitudinally between a front and a rear;
- a handle pivotally attached to the housing at a handle/housing pivot, the handle/housing pivot being at an upper front location of the housing;
- a fastener guide track disposed along the bottom of the housing;
- a striker disposed at the front of the housing including an upper striker position above the track and a lower striker position in front of the track;
- a grip body of the housing elongated front to rear below the handle;
- a grip opening of the housing elongated front to rear below the grip body;
- a power spring supported within the housing, the power spring being a torsion type including a spring coil, the spring coil being confined within the grip body;
- the power spring having a first spring arm extending forward from the spring coil to a first spring end, the first spring end linked to the striker to move with the striker, a second spring arm extending forward from the spring coil to a second spring end;
- a lever extending rearward from a lever front end adjacent to the striker to a lever rear end, the lever linked to the power spring second spring arm whereby downward motion of the lever causes the second spring arm to deflect and store energy, the lever including a position adjacent to the terminal coil winding;
- a spring coil axis extending cross a width of the grip body between the first terminal coil winding and an opposed second terminal coil winding, the second terminal coil winding including an upper winding portion spaced from the sidewall of the grip body while the first

terminal coil includes a top coil termination adjacent to an opposed wall of the grip body; and

- a rear of the spring coil includes four adjacent coil winding segments abutting each other.

18. The fastening tool of claim **17**, wherein the top coil termination contacts the opposed wall of the grip body.

19. The fastening tool of claim **17**, wherein the lever includes a lever rear end, and the lever pivots within the grip body at a location of the lever rear end substantially coincident with the spring coil axis.

20. The fastening tool of claim **19**, wherein the lever rear end is nested within the second terminal coil winding.

21. A fastening tool, comprising:

- a housing with a top, bottom and sides, the housing extending longitudinally between a front and a rear;
- a handle pivotally attached to the housing at a handle/housing pivot, the handle/housing pivot being at an upper front location of the housing;
- a fastener guide track disposed along the bottom of the housing;
- a striker disposed at the front of the housing including an upper striker position above the track and a lower striker position in front of the track;
- a grip opening of the housing having an elongated shape from front to rear;
- a torsion power spring supported within the housing having a spring coil with at least 3-1/3 turns and confined within a housing space above the grip opening;
- wherein the power spring includes a first spring arm extending forward from the spring coil and linked to the striker to move with the striker, and a second spring arm extending forward from the spring coil; and
- a lever extending rearward from a lever front end adjacent to the striker to a lever rear end, wherein the lever is linked to the power spring second spring arm, and wherein downward motion of the lever causes the second spring arm to deflect and store energy, the lever being at least partially nested within the spring coil.

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